Copernicus Global Land Operations "Cryosphere and Water"

"CGLOPS-2"

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ALGORITHM THEORETICAL BASIS DOCUMENT

LAKE WATERS

300M PRODUCT

VERSION 2.0.0

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Organisation name of lead contractor for this deliverable:

Book Captain:	Stefan Simis (PML)
Contributing Authors:	Kerstin Stelzer (BC), Dagmar Müller (BC), Nick Selmes (PML), Eirini Politi (BC)

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List of Acronyms

ATBD	Algorithm Theoretical Basis Document
BC	Brockmann Consult
CCI	Climate Change Initiative
C-GLOPS	Copernicus Global Land Operations
FTP	File Transfer Protocol
GCOS	Global Climate Observing System
GIO	GMES Initial Operations
GLWD	Global Lakes and Wetlands Database
GPT	Graph Processing Tool
LSR	Lake Surface Reflectance
LWQ	Lake Water Quality
MERIS	Medium Resolution Imaging Spectrometer on Envisat
MSI	Multispectral Instrument on Sentinel-2
NERC	National Environment Research Council (UK)
NRT	Near Real Time
NTU	Nephelometric Turbidity Unit
obs	Observation
OC	Ocean Colour
OLCI	Ocean and Land Colour Instrument on Sentinel-3
OWT	Optical Water Type
PML	Plymouth Marine Laboratory
PUM	Product User Manual
QAA	Quasi-Analytical Approach
QAR	Quality Assessment Report
rep	representative
R_w	Fully normalised water-leaving reflectance
TS	Trophic State
TSI	Trophic State Index
TSM	Total Suspended Matter
TUR	Turbidity
WGS84	World Geodetic System 1984

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1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Copernicus Global Land Service – Lake Water Quality component provides an optical characterisation of more than 4200 inland waterbodies. These waterbodies belong to the world's largest or are otherwise of specific environmental monitoring interest. The set of water bodies that are included in the observations consist for the largest part of natural lakes, however reservoirs, enclosed lagoons and wide river sections are also included. The products contain six (sets of) variables: lake water-leaving reflectance (all wavebands available after atmospheric correction), turbidity, total suspended matter concentration, chlorophyll-a concentration, a floating cyanobacteria index and a trophic state index (derived from phytoplankton biomass by proxy of chlorophyll-a). Some of these estimates now benefit from uncertainty models developed in the context of the ESA Lakes Climate Change Initiative (Lakes_cci), providing per-pixel estimates of relative uncertainty.

This document describes the 300m water quality products, version 2.0 of the processing chain. Production and delivery are over set 10-day intervals (starting the 1st, 11th and 21st day of each month) and mapped to a common global spatial grid at nominally 300m (~0.0022°) spatial resolution. The input products are Envisat MERIS and Sentinel-3 OLCI L1b data. The algorithms used to derive the input for the optical lake water products are implemented in the *Calimnos* processing chain and were tuned and validated against 13 predefined optical water types (OWT), initially in the UK NERC UK GloboLakes project, and later in the ESA Lakes_cci project. This Algorithm Theoretical Basis Document (ATBD) describes these algorithms, their evolutions and their use in the *Calimnos* processing chain.

1.2 SCOPE AND OBJECTIVES

One of the main objectives of the Copernicus programme is to provide to the scientific community and other stakeholders, including policy makers, state-of-the-art information required for their respective applications. Prior to product version 2.0.0, the Lake Water Quality (LWQ) products turbidity (TUR) and Trophic State Index (TSI) were already in pre-operational mode, whereas the water-leaving reflectance was in demonstration mode due to suspected limitations which do not ultimately affect the derived turbidity and TSI. New variables added in this version include Total Suspended Matter (TSM) concentration, chlorophyll-a concentration and a floating cyanobacteria index, and are in demonstration mode pending further review. The original products are provided for the full Envisat MERIS and Sentinel-3 OLCI (A/B) archives as versions 1.3.0 and 1.4.0. The new version enters near real-time (NRT) using OLCI. The objective of this document is to provide details on the algorithms and processing chain used for the optical LWQ products at 300m.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements and the expected performance
- Chapter 3 describes the processing chain
- Chapter 4 lists the algorithms underlying the retrieval of individual parameters

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: Call for tenders JRC/2020/OP/3213 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2020)7715271, dated 17/12/2020

AD4: Call for tenders JRC/IPR/2023/CPN/2325 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2023)6699927, dated 03/10/2023

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service.

1.4.3 Output

Document ID	Descriptor
CGLOPS2_PUM_LWQ300_v2.0.0_I2.01	Product User Manual summarizing all information about the Lake Water Quality products, demonstration and pre-operational products, historic and NRT data
CGLOPS2_QAR_LWQ300_v2.0.0_I2.01	Report describing the results of the scientific quality assessment of the Lake Water Quality products, demonstration and pre-operational products, historic and NRT data

1.4.4 External documents (if any)

N/A

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2 REVIEW OF USER REQUIREMENTS

According to the applicable document [AD2], the user requirements relevant for Lake Water Products v2.0 are:

Definition:

The Lake Water Products are lake turbidity (TUR) in NTU, total suspended matter (TSM) in g/m^3, Trophic State Index (TSI), chlorophyll-a concentration (Chl-a) in g/m^3, floating cyanobacteria index and lake surface reflectance spectra (LSR, given in terms of fully normalised water reflectance R_w; a dimensionless quantity) from which TUR, TSM, Chl-a, TSI and the floating cyanobacteria index are derived. In addition, the mean of relative uncertainty of Chl-a and TSM in the observation period, and the relative uncertainty and unbiased uncertainty in LSR at reflectance peak are included. The products shall be provided as 10-day averages of the respective parameters, except for LSR for which the observation that is statistically most representative for the observed time period is given. The reason for the latter is that an average of each LSR waveband may yield a physically inaccurate spectrum of reflectance and would, thus, not be useful to feed into alternative algorithms for the retrieval of optical water quality parameters. Turbidity and TSM are key indicators of water clarity. Turbidity quantifies the haziness of the water and acts as an indicator of underwater light availability. TSM quantifies the concentration of suspended particles in the water body. TSI, as defined by Carlson (1977), refers to the degree at which organic matter accumulates in the water body and is most commonly used in relation to monitoring eutrophication. Chl-a is the precursor product to Trophic State Index and is commonly used as a proxy to phytoplankton biomass. The cyanobacteria risk index indicates the detection of cyanobacteria surface blooms in the water body.

Geometric properties:

The pixel size of the baseline dataset shall be provided at resolutions of 100m and/or 300m. The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view. Pixel coordinates shall be given for the pixel centre.

Geographical coverage:

Global window

The initial window definition is aligned to the global datasets produced during the GMES Initial Operations (GIO) phase for the most widely used output data. The following output specifications are further optimised with respect to the requirements:

- pixel size at 300m: 0.25/112°
- global window coordinates: UL: 180°W-90°N, BR: 180°E, 90°S
- global grid size at 300m: 161280 columns, 80640 lines

Continental windows:

Continental windows may be asked by the contracting authority to satisfy specific user needs. Wherever applicable, continental windows shall be drawn from global output data.

Accuracy requirements:

Lake Water Reflectance is an Essential Climate Variable (GCOS 244, 2022) with an associated accuracy requirement threshold of 30% (breakthrough: 20% and target: 10%) at the reflectance peak (GCOS 245, 2022). Turbidity has an uncertainty threshold of 50% (goal: 35%) and TS has an uncertainty threshold of 0.3 in log10 or 50% [AD4].

Temporal Definition

As a baseline, the biophysical parameters are computed by and representative of *dekads*, i.e. periods of nominally ten and varying from eight to eleven days, defined to start on days 1, 11 and 21 of each month of the year.

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3 METHODOLOGY DESCRIPTION: PROCESSING CHAIN

3.1 OVERVIEW

CGLOPS Lake Water processing uses the *Calimnos* processing chain. *Calimnos* was initially developed in UK NERC GloboLakes with input from the ESA Diversity-II project (2015) and has been further developed with core collaboration between Plymouth Marine Laboratory (PML), University of Stirling, Brockmann Consult (BC) and HYGEOS in the Copernicus Land Monitoring Service, H2020 TAPAS, H2020 EOMORES, and ESA Lakes_cci. The chain can be used to process archived Envisat MERIS, Sentinel-3 OLCI and Sentinel-2 MSI data for operational processing. This document refers to 300m products based on OLCI generated using *Calimnos* v2.0.

Calimnos combines into a single processing chain: data discovery, spatial resampling and subsetting by target area (individual water bodies, image granule or any other region of interest), radiometric and atmospheric corrections, pixel identification (land/cloud/water/ice), optical water type classification, individual algorithms (per parameter and water type), weighted blending of results, optional reprojection and spatiotemporal aggregation.

3.2 THE RETRIEVAL METHODOLOGY

3.2.1 Outline

A schematic overview of *Calimnos* is given in Figure 1. The main processing stages include:

- Data discovery. Newly downloaded L1b satellite passes are entered into a geospatial database. Target regions are similarly specified in a geospatial database and satellite products which overlap any of the target regions are queued for processing. In the context of re-processing, any duplicate passes are removed. The procedure relies on in-house Python scripts and PostgreSQL database functionality.
- Subsetting. For best processing performance, satellite passes are subset to bounding boxes around each target area. The target area is defined as the maximum water extent observed in ESA CCI Land Cover product v4. The subsetting routine is part of the SNAP toolbox, called through the Graph Processing Tool (GPT).
- Radiometric corrections. Any radiometric corrections defined by ESA and/or EUMETSAT following the release of the data are applied to the L1b imagery before submitting the data to atmospheric correction. For MERIS the data from 3rd reprocessing and subsequently published radiometric corrections are used as part of the SNAP toolboxes using GPT. The 4th reprocessing of MERIS data will be used from version 3.0 of the lake water quality products.
- Pixel identification. The *IdePix* (Identification of pixel categories) neural network routine is applied for initial pixel identification as water, land, cloud/haze, cloud shadow or snow/ice.

IdePix is called through SNAP using the GPT. Pixel identification masks are stored for later masking of invalid (non-water) pixels.

- Atmospheric correction. Polymer is applied to the corrected L1b data and yields water-leaving reflectance wavebands. The outputs are fully normalised water-leaving reflectance (R_w). Polymer is called using a function wrapper in Python.
- Optical water type classification. The optical water type (OWT) classification developed in the UK NERC GloboLakes project (Spyrakos et al. 2018) is applied to each pixel to determine the similarity of the observed water-leaving reflectance spectrum to thirteen known OWTs. Two additional OWTs are used for identifying pixels contaminated by land adjacency (Jiang et al. 2023).
- Algorithm mapping and blending. For each OWT a best-performing algorithm (see section 4) has been selected and tuned against global datasets of in situ measurements. For CGLOPS the best performing algorithm corresponding to the OWT of a given pixel is mapped for turbidity, TSM, and chlorophyll-*a* as a precursor for trophic state. These final products are derived as described in Section 4.
- Floating Cyanobacteria algorithm: Detection of surface blooms by applying the Maximum Peak Height (MPH) algorithm (Mathews and Odermatt 2015) adapted to use top-of-atmosphere (TOA) reflectance instead of Bottom-of-Rayleigh reflectance (BRR). The thresholds of the MPH were adapted to generate the same cyanobacteria surface bloom coverage from TOA signals compared to BRR, which reduces computation time. The algorithm differentiates between floating cyanobacteria and floating vegetation using spectral indices.
- Aggregation. 10-day aggregation is done on a per-lake basis after applying masks to select data for water pixels. The reflectance products of the Polymer atmospheric correction are inspected over the 10-day period and for each pixel the most representative spectrum is chosen (see Section 4). The aggregated products which contain the LSR, TSI, TSM, Chl-a and TUR parameters are then mapped to a global grid at 0.0022° (nominally 300m).

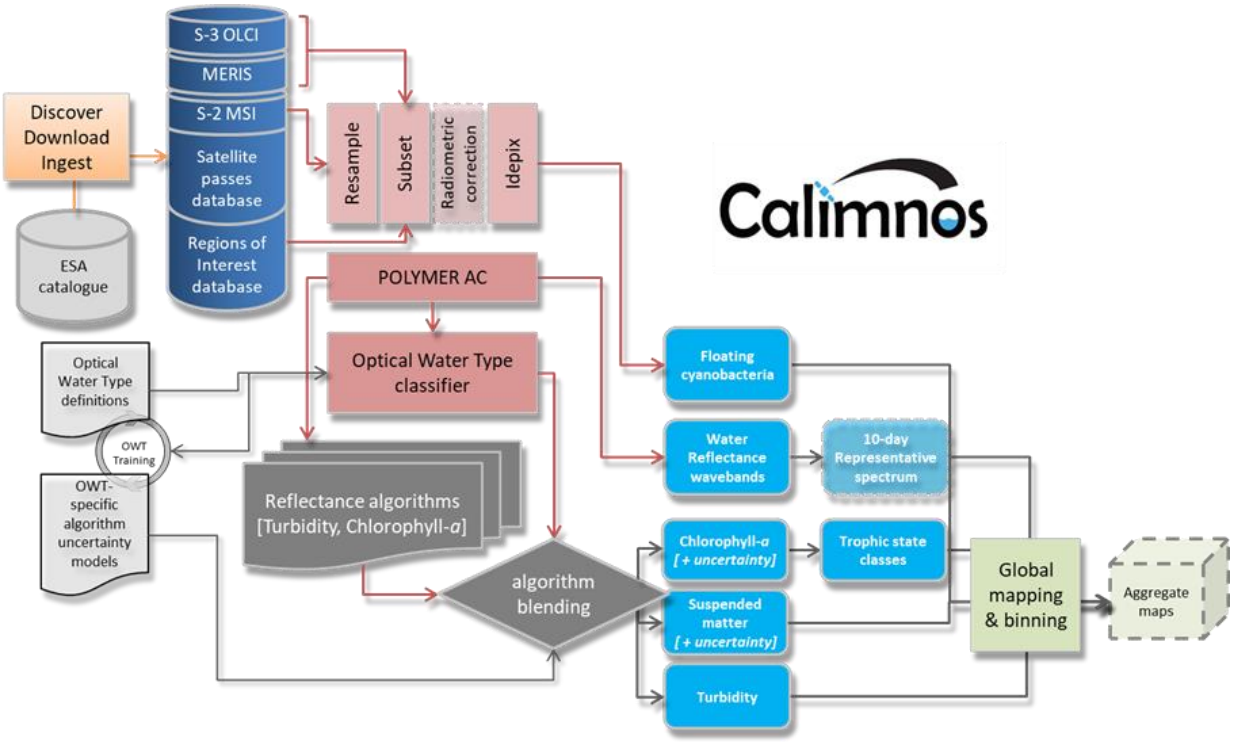


Figure 1. Schematic overview of the *Calimnos* processing chain v2.0 for 300m product generation

3.2.2 Basic underlying assumptions

The processing chain is subject to several sources of uncertainty including ambiguity in classification and algorithm calibration and these appear in various stages of the processing chain. The validity of the resulting products is subject to validation and is ultimately based on a number of theoretical and practical assumptions on the performance of individual processing elements:

Assumptions addressed in product validation (see Quality Assessment Report);

- *IdePix* can differentiate adequately between pixels containing water and any of the following conditions: mixed land/water, cloud, cloud shadow, ice, snow, and haze.
- Polymer successfully retrieves the shape and amplitude of water-leaving reflectance.
- Tuning of reflectance algorithms for chlorophyll-a, suspended matter, and turbidity is adequate for each water type.
- The water-leaving reflectance spectra can be assigned a membership score for at least one known optical water type.
- The tuned reflectance algorithms for chlorophyll-a, suspended matter and turbidity yield valid results over the range of reflectance observed over water types to which they have been assigned.
- Selection of the most representative spectrum from observations obtained over 10 days uses statistical metrics that result in a consistent pattern for the whole water body.
- The translation of chlorophyll-a to Trophic State Index is adequate and consistent between water types.

Fundamental assumptions that are currently not independently addressed:

- The 13 optical water types defined in UK NERC GloboLakes sufficiently capture the natural optical diversity of inland waterbodies. More in situ observation data are needed to assess current gaps.

The most crucial assumption of the *Calimnos* processing chain is that optical water types which have been defined from a large set of in situ data from optically complex waters (lakes, reservoirs, lagoons, estuaries, and coastal areas) can be assigned successfully to each satellite observation (pixel) containing open water. In practice, the assignment to an optical water type can fail due to various reasons, the most prominent being failure to identify the pixel as water (or as influenced by cloud or ice) by *IdePix*, or failure in the atmospheric correction yielding a reflectance spectrum that is incompatible with any of the algorithms used downstream. Misclassification of a given image pixel will normally be associated with a low classification score giving rise to caution in using the resulting products for that pixel. Solutions for misclassification and improvement of atmospheric correction are continuously being explored.

3.2.3 Related and previous applications

The most recent version of the Calimnos NRT processing chain is 2.0 (Table 1). This version differs from the previous version 1.4.0 in several ways. Firstly, the atmospheric correction processor is updated from Polymer v4.12 to v4.14. The out-of-bounds flag from Polymer is applied. Additional variables now included are chlorophyll-*a*, Total Suspended Matter, and the floating cyanobacteria index. Previously, TSM was calculated and converted to Turbidity, whereas TSM is now included as a separate product while Turbidity is calculated independently using the Nechad algorithm (Nechad et al. 2010, 2016; section 4.2.5). Per-pixel uncertainty variables are supplied for chlorophyll-a, TSM, and water-leaving reflectance.

For OLCI, Polymer v4.12 and v4.14 provide equivalent outputs. Therefore, reflectance and trophic state products found in Copernicus products v1.4.0 and v2.0 are fully compatible. Turbidity in v1.4.0 was calculated by applying a scaling factor to TSM, whereas in v2.0 a separate algorithm is used. Turbidity equivalent to v1.4.0 can be obtained in v2.0 by applying the following scaling factor to TSM: Turbidity = TSM * 1.17 (see section 4.2.4).

Older product versions

The previous version of the 300m LWQ product was v1.4.0, which differed from version 1.3.0 by an updated version of *IdePix* with improved lake ice detection. Products generated using v1.3.0 used a static, and, therefore, likely incomplete, ice mask in the period 2016-2019 (OLCI sensors), whereas ice masking for the MERIS sensor (2002-2012) performed nominally. These products are currently available for the reprocessed period 2002-2019, whereas v1.4.0 was introduced in June 2020. The period Jan-Jun 2020 was filled using v1.4.0.

The first release for the Copernicus Global Land service used *Calimnos* v1.1.0 for reprocessing of the MERIS archive. *Calimnos* v1.0.4 was used to produce the first global adaptive water quality dataset in the UK NERC GloboLakes project to relate water quality changes to environmental

change in 1000 lakes globally. Compared to GloboLakes processing, the temporal aggregation procedure was improved to select Lake Surface Reflectance as the statistically most representative reflectance spectrum for the 10-day period, and to aggregate the Trophic State Index from chlorophyll-a concentration. The selection of the most representative spectrum is described in 3.2.1 Aggregation.

The selection of lakes has also evolved. Earlier product versions included an initial set of 1000 lakes, which covered more than 50 % of the global surface area of inland water bodies (1,409,030 km^2, not counting the Caspian Sea which is also included in the service). Since version 1.3.0 of the processing chain, the set of lakes has been extended to 4,265 waterbodies to a total of 2,166,023 km^2. The included waterbodies span a surface size range of 0.11 to 379,446 km^2.

3.2.4 Planned future versions

A future version 3.0 is anticipated for 2025, dependent on the development of a new set of algorithms calibrated to use the next iteration of Polymer which is designed to give improved results in very turbid water. The anticipated LWQ product v3.0 will include updated water-leaving reflectance variables, and derived products returned for the updated inputs. The various versions of past, present, and future products are summarised in Table 1.

Table 1. Summary of Lake Water Quality product versions. Polymer versions marked with * are equivalent for OLCI

Product version	*Idepix* version	Polymer version	Other changes from previous version
1.2	6.x		
1.3	7.0.4	4.12*	OWT matching uses spectral angle not mahalanobis distance
1.4	7.0.5	4.12*	Improved ice masking in *IdePix*
2.0	7.0.5	4.14*	Cyanobacteria product Separate turbidity and TSM algorithms Chl-a supplied as well as trophic state Per-pixel uncertainty provided (Chl-a, TSM) Additional mask bands Polymer out-of-bounds masking applied
3.0	*9.x*	*4.15 or higher*	*Major Polymer version change* *Algorithms retuned for Polymer 4.15 or higher* *Major update for IdePix*

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3.2.5 Alternative methodologies currently in use

There are different algorithms for atmospheric correction and biochemical substance estimates available in the aquatic remote sensing community. The approach to combine multiple algorithms depending on their optical water class for inland waters was also performed within the GLaSS project (http://www.glass-project.eu/; all documents available here: http://www.glass-project.eu/downloads/) and the Diversity-II (Diversity-II 2015; http://www.diversity2.info/) project. Algorithms are generally calibrated and validated against a limited number of lakes. The approach used in *Calimnos*, in contrast, enables the processing of globally distributed and optically very diverse lakes.

Algorithm mapping and blending depending on optical water type membership is similar to the strategy used to produce the ESA Climate Change Initiative Ocean Colour datasets, although *Calimnos* compares reflectance spectra based on shape rather than magnitude as this is deemed more suitable for water types with dynamic substance concentrations.

3.2.6 Input data

Satellite input data:

- Envisat MERIS L1b, 3rd reprocessing (4th reprocessing from LWQ v3.0 onward), full resolution, downloaded from ESA.
- Non-Time-Critical Sentinel-3 OLCI L1b data from A and B platforms, downloaded from multiple sources (SAFE format).

Lake water boundaries:

- maximum water extent observed in ESA CCI Land Cover (v4.0) at 150-m resolution
- polygons generated (including manual inspection) at PML (latest release: https://doi.org/10.5281/zenodo.835851; or latest development version: https://github.com/pmlrsg/lake-polygons-PML/tree/master/CLMS4k).

Polymer ancillary data:

- No ancillary data are required when the Non-Time-Critical L1b data are used.

Optical water type definitions (spectral means):

- UK NERC GloboLakes project (Spyrakos et al. 2018).

3.2.7 Output product

Global, regional, or water-body specific files in NetCDF4 format, mapped to a global grid and including dimensions latitude, longitude (WGS84 projection), and time (days since 1 January 1970). Each file includes the variables indicated in Table 2.

Table 2. Variables included in the products

band name	data type	description
num_obs	int	Number of observations (including non-water observations)
first_obs	float	Number of days from the start of the sensing period until the first cloud-free observation
last_obs	float	Number of days from the start of the sensing period until the last cloud-free observation
turbidity_mean	float	Mean turbidity in the observation period
chla_mean	float	Mean of chlorophyll a concentration in the observation period
chla_uncertainty	float	Mean of relative uncertainty of chlorophyll a concentration in the observation period
tsm_mean	float	Mean of total suspended matter in the observation period
tsm_uncertainty	float	Mean of relative uncertainty of total suspended matter in the observation period
R_w400_rep (OLCI)	float	Fully normalised water-leaving reflectance corresponding to the most representative spectrum for the observation period, at the waveband indicated in the band name. Band names indicated as 'OLCI' in the left column only occur with that sensor.
R_w412_rep	float	
R_w443_rep	float	
R_w490_rep	float	
R_w510_rep	float	
R_w560_rep	float	
R_w620_rep	float	
R_w665_rep	float	
R_w674_rep (OLCI)	float	
R_w681_rep	float	
R_w709_rep	float	
R_w754_rep	float	

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band name	data type	description
R_w779_rep	float	
R_w865_rep	float	
R_w885_rep	float	
R_w900_rep	float	
R_w1020_rep (OLCI)	float	
R_w_relative_uncertainty	float	Relative uncertainty in fully normalised water-leaving reflectance at reflectance peak
R_w_relative_uncertainty_unbiased	float	Relative unbiased uncertainty in fully normalised water-leaving reflectance at reflectance peak
R_wDayNum_rep	float	Age, in days, after the start of the observation period, of the most representative water-leaving reflectance spectrum
trophic_state_index	float	Trophic state index (TSI) obtained from chlorophyll-a observations, averaged over the observation period. The TSI has a value of 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, or 100 corresponding to the classes given in Carlson (1977), where 0-40 are oligotrophic, 40-60 mesotrophic, 60-80 eutrophic, and 80-100 hypereutrophic (Table 4).
floating_cyanobacteria	float	Floating cyanobacteria index, with a range of values from 0 to 1 indicating probability of cyanobacteria presence – the closer the value is to 1, the higher the probability
lat	float	Latitude in decimal degrees
lon	float	Longitude in decimal degrees
time	float	Time in days since 1-1-1970
crs	string	Coordinate reference system information
quality_flags	int	Flag band showing why input observations to a pixel have been included. Flags are l1_invalid, atmospheric_correction_failure, cloud, land, land_contaminated, snow_ice, bright_pixels, turbidity_invalid, chla_invalid, tsm_invalid, floating_cyanobacteria,

band name	data type	description
		floating_macrophytes_or_mixed_pixel

3.2.8 Methodology

The main processing steps are given in the outline (section 3.2.1) above. Additional detail of the algorithms underlying pre-processing and atmospheric correction is included below. Section 4 details the reflectance algorithms used to retrieve chlorophyll-a and total suspended matter concentrations for each optical water type.

Satellite passes are always first subset to match the area of interest (a water body polygon) prior to further processing to reduce computation time. The area of interest is defined as the maximum water extent reported in the ESA CCI Land Cover project (v4.0). Ephemeral or dynamic water bodies will likely contain land pixels within this area of interest. These are recognised as part of the pixel identification step and only pixels identified as (open) water are processed further.

Details on the individual modules and algorithms, with links to documentation where available, are given below. Wherever SNAP is mentioned this refers to SNAP v7. All SNAP modules will be updated to SNAP v10 or higher with LWQ version v3.0.

Pre-processing:

- SNAP Subset operator.
- MERIS L1b Radiometric Correction v1.2, comprising optional smile correction (not used in combination with Polymer atmospheric correction) following the MERIS ATBD and a radiometric equalisation following Bouvet and Ramoino (2010).
- OLCI-A calibration gains for Polymer (provided by ESA CCI Ocean Colour) are applied; OLCI-B calibration gains will be applied from LWQ version 3.0.

Pixel identification:

- Pixel identification and classification over water (and land), with an algorithm developed within the CoastColour and ESA CCI Ocean Colour projects, uses *IdePix*:
    - MERIS: SNAP IdePix-Water Module v7, Operator Idepix.Meris
    - OLCI: SNAP IdePix-Sentinel3 Module v7.0.5, Operator Idepix.Olci

Atmospheric correction:

- Polymer according to Steinmetz et al. (2011), updated in Steinmetz (2016 and 2018), parameterized to use the Park and Ruddick (2005) bidirectional reflectance distribution function, and operating only on pixels identified as water by the *IdePix* module:
    - MERIS: Polymer v4.14
    - OLCI: Polymer v4.14

Per-pixel uncertainty:

- Uncertainty is calculated per pixel for Chl-a and TSM according to Liu et al. (2021) based on the relation between satellite vs in situ residuals and Optical Water Type similarity (see also ESA Lakes_cci 2022). The per-pixel uncertainty for water-leaving reflectance propagates uncertainty from regression fits between satellite and in situ observations, without discriminating OWTs. This is due to scarcity of in situ radiometric observations.

Optical Water Type (OWT) classification:

- Implemented in Python at PML based on the work of Moore et al. (2001) and equivalent software developed for ESA CCI. The algorithm uses standardised Optical Water Type spectra defined in the UK NERC GloboLakes project, by the University of Stirling (Spyrakos et al. 2018). In contrast to OWT mapping used in some previous projects, we apply the spectral angle (Kruse et al. 1993) rather than Mahalonobis distance as metric for similarity between spectra, as it allows for better distinction between the clearest waters (where covariance, on which Mahalonobis distance relies, is very low in longer wavebands) and more turbid water types.

Water constituent algorithm mapping:

- Water constituent retrieval algorithms tuned to each OWT in the UK NERC GloboLakes project (Spyrakos et al. 2018, Neil et al. 2019), are mapped to each individual pixel according to the OWT with the three highest classification scores for that pixel. The algorithm results corresponding to those three OWTs are averaged using the membership score as weighting. Algorithms giving invalid results are not included in the weighted average. The weighting factors are scaled between 0 and 1, where 1 is the highest OWT membership score and 0 is the next OWT score that is not included in averaging (i.e. the fourth). This procedure is used to derive maps of total suspended matter, turbidity, and chlorophyll-a. TSI is then calculated from Chl-a. The underlying algorithms are listed in Sections 4.2.3 and 4.2.4.

Floating cyanobacteria/macrophytes indicators:

- Detection of surface blooms based on the methodology used for the Maximum Peak Height (MPH) algorithm in Matthews and Odermatt (2015). Differentiation between floating cyanobacteria and floating vegetation is done by using spectral indices.

Aggregation:

- The aggregation step performs spatial and temporal aggregation of pixel values into cells ('bins') of a planetary grid over a lake. During the lake aggregation, invalid pixels are masked before averaging. Mean values in each spatiotemporal bin are derived for the remaining observations. Invalid pixels are recognised based on L1 product flags (L1.INVALID), the *IdePix* cloud, ice, haze, and land identification flags, Polymer out-of-bounds flag, and value limits for individual products (see the invalid flag definitions for chl-a, TSM and turbidity in Table 3).
- Most representative spectrum selection. The algorithm has been developed by PML and BC for use in the Copernicus Global Land Service. For each pixel, water-leaving

reflectance spectra are compared to the median spectrum using the root-mean-square (RMS) difference across all wavebands. The spectrum with the lowest RMS difference is selected as most representative for the observation period.

- 10-day aggregation is done on a per-lake basis after applying masks to select data for water pixels. The aggregated products which contain the LWQ variables (reflectance, trophic state, turbidity, TSM, chl-a and cyanobacteria risk index) are then mapped to the global grid.

Quality flags:

For the biogeophysical variables, each pixel in the final products is an aggregation of many observations. The number of underlying observations will vary as some inputs are masked due to insufficient quality, contamination by cloud or ice and other factors. The *num_obs* variable gives the total number of input observations to a pixel before quality control. A *quality_flag* is provided which indicates what masking has removed input observations from the aggregated final output. Flags raised in this variable do not indicate a degradation in the quality of the final aggregated pixel, but a reduction in the number of input observations which may reduce the representativeness of the result over the aggregation period. Table 3 shows the meaning of these quality flags.

Table 3. Quality flag definitions

Flag name	Definition	Explanation
bright_pixels	Rw412 > 0.4 or Rw865 > 0.4	One or more input observations removed as anomalously bright
cloud	pixel_classif_flags.IDEPIX_CLOUD or pixel_classif_flags.IDEPIX_CLOUD_BUFFER or pixel_classif_flags.IDEPIX_CLOUD_SHADOW or pixel_classif_flags.IDEPIX_CLOUD_AMBIGUOUS or pixel_classif_flags.IDEPIX_CLOUD_SURE	One or more input observations removed as were flagged by *Idepix* as being affected by cloud
snow_ice	pixel_classif_flags.SNOW_ICE	One or more input observations removed as flagged by *Idepix* as snow or ice
l1_invalid	l1_flags.invalid	One or more input observations removed as instrument data missing or invalid,

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Flag name	Definition	Explanation
		flagged in source data
land	land_water_fraction < 100	One or more input observations removed as some percentage of the pixel is land
land_contaminated	(owt_14 + owt_15>1.59) and ((owt_dominant_class==14) or (owt_dominant_class==15))	One or more input observations likely contaminated by adjacency effect, but not removed (source variables for this expression not supplied)
atmospheric_correction_failure	Polymer bitmask: OUT_OF_BOUNDS	One or more input observations removed as Polymer bio-optical model reached its concentration limits
turbidity_invalid	turbidity > 100000 or turbidity < 0	One or more input observations removed as turbidity value likely unrealistic (suspected unmasked ice)
chla_invalid	chla < 0 or Chla > 10000	One or more input observations removed as chl-a value likely unrealistic
tsm_invalid	tsm < 0 or tsm > 10000	One or more input observations removed as TSM value likely unrealistic
floating_cyanobacteria	floating_cyanobacteria > 0.4 and not (l1_invalid, cloud, land, land_contaminated)	Floating cyanobacteria likely in one or more input observations
floating_macrophytes_or_mixed_pixel	floating_macrophyte_probability (not supplied) > 0.5 and not (l1_invalid, cloud, land, land_contaminated)	One or more input observations likely contained floating macrophytes or was a

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Flag name	Definition	Explanation
		mixed pixel

3.2.9 Limitations and known issues

Despite a lack of sufficient in situ reference data, systematic underestimation of water-leaving reflectance is apparent in the current product version, particularly for highly turbid lakes, when compared to in situ measurements. This bias has been investigated and attempts have been made to adjust the water reflectance model used in the Polymer algorithm and make it more representative of highly reflective lake waters. An updated version of Polymer (v4.15) addressing this issue and subsequent algorithm calibration is undergoing and is expected to feature in the service under product v3.0.

Calimnos assigns the most appropriate algorithms for chl-a, TSM and turbidity based on the per-pixel similarity to known optical water types. In practice, atmospheric correction may have systematic errors for some water types, leading to low membership scores. This, in turn, implies that suboptimal reflectance algorithms for chlorophyll-a and suspended matter may be selected. Solutions for this issue are being explored. The inclusion of per-pixel uncertainty is one step in this direction. However, these uncertainty estimates propagate from global (per-OWT) algorithm performance and may not accurately reflect specific observation uncertainty. For example, observations near land under atmospheric haze are likely to be less reliable than observations in open bodies of water under clear atmospheric conditions.

If the pixel identification and thus the flagging of erroneous pixels does not work properly, e.g. at cloud borders, thin clouds or lake ice coverage, the water-leaving reflectance and subsequent water quality parameter retrieval may fail. Users interested in lake averages rather than maps are advised to mask a buffer of one or more pixels from shorelines, where such effects are most difficult to control and flag. An upgrade for the lake ice detection was introduced with version 1.4.0 (in production since June 2020). Version 1.3.0 data (before June 2020) are affected by less performant ice detection, although a static ice mask was applied to mask locations where ice is likely to occur.

The spatial blending of different algorithms and/or the averaging of different, very inhomogeneous days under partial cloud cover can lead to visually inconsistent maps. Lakes may show patchy patterns where coverage on different days of observation was incomplete, and if optical conditions in the lake changed over this period. This is considered normal behaviour for the dekadal product.

The radiometric correction of OLCI on Sentinel 3A is implemented based on system vicarious calibration results. OLCI-B currently follows pre-flight calibrations while results from the tandem flights are pending. OLCI-B gains will be applied from LWQ product version 3.0.

3.3 QUALITY ASSESSMENT

The quality assessment is described in detail in chapter 4.3.

3.4 RISK OF FAILURE AND MITIGATION MEASURES

The historic products are based on archived MERIS products. Therefore, a failure of input data is not expected. Processing of NRT data starts at L1 and failures in timely provision are possible and communicated via the service helpdesk.

The processing chain is version controlled and backed-up at PML.

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4 METHODOLOGY DESCRIPTION: RETRIEVAL ALGORITHMS

4.1 OVERVIEW

This section provides additional detail and references to published work for the algorithms implemented in the *Calimnos v2.0* processing chain.

4.2 THE RETRIEVAL ALGORITHMS

4.2.1 Pre-processing

For the pre-processing, methods are used that have been applied in many different production chains and well discussed in other documents (e.g., Diversity-II ATBD, CoastColour ATBD, GlobWetland Africa ATBD). The steps to be applied are:

1. Geometric correction / optimisation
2. Coherent Noise Equalisation
3. Smile correction
4. Cloud and cloud shadow detection by applying *IdePix*

The standard MERIS **geometric correction** used for the 3rd reprocessing needs an optimisation step, which is performed by the AMORGOS software developed by ACRI-ST and provided by ESA (http://earth.esa.int/services/amorgos/download/Amorgos_ICD-SUM_3.0a.pdf and http://earth.esa.int/services_/amorgos/download/Amorgos_STD_i3r0p1.pdf). It includes a precise orbit determination, instrument pointing and performs an ortho-rectification. Improvement in geolocation is documented in Bicheron et al. (2011). For *Calimnos*, the AMORGOS geometric correction is held alongside the original L1b product and patched in at the end of L2 processing. The resulting accuracy is improved by AMORGOS and is specified in Bicheron et al. (2011) to be better than 70m for MERIS full resolution (FR) pixels. These additional steps will not be required for the 4th reprocessing of MERIS.

For OLCI, no further processing step for improving the geolocation position is required. The accuracy is given by ESA to be 0.2-0.7 pixels for processing version 2.23 and <0.1 pixel for processing version 2.29.

The **coherent noise equalisation** is reducing detector-to-detector and camera-to-camera systematic radiometric differences and results into a diminution of the vertical stripping observed on MERIS L1b products following the algorithm developed by Bouvet and Ramoino (2010); coefficients for each detector line are retrieved from MERIS archive over ice. Another step for radiometric improvement is the **smile correction.** It corrects for the small variations of the spectral wavelength of each pixel along the image by the estimation of the reflectance spectral slope from the measurements in two neighbouring bands. The reflectance along this slope is interpolated from the actual pixel wavelength to the nominal wavelength of the considered band (Bourg et al. 2008, Delwart et al. 2005). For the avoidance of doubt, the smile correction is not used in the *Calimnos*

processing chain which implements Polymer, as the latter follows a pixel-by-pixel model of the spectral distribution of each waveband.

The cloud detection function of the *IdePix* tool developed by BC was used in several processing chains, e.g., those used in CoastColour L1P and ESA Diversity II (2015). Many steps of the *IdePix* process are included in the MERIS 4th reprocessing as standard by ESA. Due to the good performance of *IdePix* cloud screening in these applications, it is also selected for the *Calimnos* processing. *IdePix* is based on a cloud probability derived from a neural net which has been trained with >60,000 manually classified pixels and which is combined with a number of additional tests on, for example, brightness, whiteness, and glint. The *IdePix* CLOUD mask contains the CLOUD_SURE and the CLOUD_AMBIGUOUS flag, which identify opaque clouds as well as semi-transparent clouds. After clouds have been identified, a buffer can be defined to provide a safety margin along cloud borders. This buffer radius (in pixels) can be parameterised and it is mainly used with a buffer of 1 or 2 pixels.

Validation is performed by applying the *PixBox* Validation, a procedure where manually selected pixels are categorised to different categories and characterised with expert knowledge, e.g., to clear land, clear water, totally cloudy, semi-transparent cloud, cloud shadow, snow/ice, *etc.* A set of 17,000 MERIS FR pixels was collected in the scope of the CoastColour project, and detailed validation results are provided in the corresponding report (Ruescas et al., 2014).

The retrieval of water quality parameters is also strongly influenced by the occurrence of cloud shadow, which needs to be identified and eliminated from further processing. Potential cloud shadow areas are identified by the geometry of the sun angle, viewing angle, cloud height and cloud bottom (base). The cloud height is retrieved by either the pressure or the temperature, but if this information is missing (not all sensors offer the respective bands), a maximum cloud height needs to be defined. The most difficult prediction is the height of the cloud base as it is not seen by the sensor. In *IdePix* it is defined as the minimum cloud height detected within the respective cloud minus an offset. The basis of good cloud shadow detection is accurate cloud detection. Validation of the cloud shadow detection is done by visual inspection of different images under different conditions (cloud types and geometries).

4.2.2 Atmospheric Correction (Lake Water Reflectance)

Polymer is developed by HYGEOS; version 4.14 is the result of further development of an atmospheric correction processor for clear ocean (case-1) waters able to deal with sunglint (Steinmetz et al. 2011), making it increasingly suitable to deal with a variety of optically complex waters, including inland waters. Polymer applies a spectral optimisation based on a bio-optical model in conjunction with radiative transfer models to separate atmospheric (including glint) and water reflectance. The principle of the algorithm is a spectral matching method using a polynomial function to model the spectral reflectance of the atmosphere and sunglint, and a bio-optical forward reflectance model for the water part. The algorithm uses the full set of wavebands available (user-configurable) as opposed to alternative ocean-colour methods that primarily extrapolate from near infra-red bands. The output is fully normalised water-leaving reflectance (R_w, dimensionless).

4.2.3 Trophic State Index

The Trophic State Index is obtained from chlorophyll-*a* according to Carlson (1977), who defined a scale comparing TSI and chlorophyll-a concentration (Table 4) as well as phosphorus concentration and Secchi disk depth, on which the TSI product is based.

Table 4. Trophic state index and related chlorophyll-a concentration classes

Trophic classification	Trophic State Index	Chl-a (mg/m³)
Oligotrophic	0	0 - 0.04
	10	0.04 - 0.12
	20	0.12 - 0.34
	30	0.34 - 0.94
Mesotrophic	40	0.94 - 2.6
	50	2.6 - 6.4
Eutrophic	60	6.4 - 20
	70	20 - 56
Hypereutrophic	80	56 - 154
	90	154 - 427
	100	427 - 1183

The Trophic State Index uses satellite-derived chlorophyll-a as input, which is retrieved based on different algorithms, depending on the OWT (Table 5). The Trophic State Index is labelled with the upper boundary of the range of trophic state (TS) values it represents, e.g. a TSI of 10 represents $0 \geq TS < 10$.

Table 5. Algorithm selection and optimisation strategy for chl-a retrieval per optical water type (OWT)

OWT number	Algorithm source	Algorithm optimisation
3, 9, 10, 13	OC2 oceancolor.gsfc.nasa.gov/cms/atbd/chlor_a	Empirical re-tuning of algorithm parameters based on GloboLakes calibration against the *Limnades* database, specific to each optical water type (Neil et al. 2019).
2, 8, 11, 12	708/665 empirical band ratio based on Gilerson et al. (2010)	
1, 4, 5, 6	Semi-analytical NIR-Red band algorithm for MERIS based on Gons et al. (2005).	
7	Adapted QAA algorithm according to Mishra et al. (2014) based on Lee at al. (2002)	

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The OWTs in the first category (3, 9, 10, 13) use the Ocean Colour Chl-a algorithm design based on a blue-to-green band ratio. The algorithm parameters were tuned in UK NERC GloboLakes to accommodate for the range of inland waters encountered in the *Limnades* database and to account for any errors in the retrieval of water-leaving reflectance from Polymer. This whole-chain validation is applied to each algorithm in *Calimnos*, recognising the fact that the most prominent error in inland water quality retrieval is the atmospheric correction step. It is noted that validation data for relatively clear inland waters are still scarce.

The remaining algorithm groups are variations of empirical and semi-analytical retrieval methods that focus on interpreting the red to near-infrared part of the spectrum. These methods are the most successful approach to retrieve chlorophyll-a concentration over a range of turbidity and trophic levels, because the red part of the spectrum is least influenced by overlapping absorption signatures of dissolved substances and other phytoplankton pigments.

The Gilerson et al. (2010) algorithm is an empirically tuned ratio of wavebands 708 and 665nm. The tuning of the algorithm is revised in *Calimnos*, by calibrating against the *Limnades* database (for OWTs 2, 8, 11, and 12) which is much larger than the dataset used in the original publication. The methodology for tuning the algorithms using in situ data is described in Neil et al. (2019). The procedure used here is the same, except atmospherically corrected satellite-derived reflectance was used as input to the tuning rather than in situ reflectance data.

The semi-analytical Gons et al. (2005) algorithm (tuned to OWTs 1, 4, 5, 6) uses the same band ratio and retrieves the backscattering coefficient from the 778nm band through inversion of reflectance into the inherent optical properties using absorption by water as a constant. Subsequently, absorption at 665nm is analytically retrieved, again through inversion but assuming backscattering is spectrally neutral. Empirical tuning of this algorithm (as above) is restricted to adjusting the slope of the backscattering coefficient and a chlorophyll-a specific absorption coefficient determined from eutrophic inland waters.

The Mishra et al. (2014) implementation of the Quasi-Analytical Approach (QAA) is very similar to the analytical inversion of Gons et al. (2005) but with a slightly different set of input bands and subsequently applied tuning parameters. In *Calimnos* it is exclusively used on OWT 7, following tuning steps as described above.

Ultimately, TSI is produced from the blended chlorophyll-a result according to Carlson (1977):

$$TSI(Chl) = 10\left(6 - \frac{2.04 - 0.68 \ln Chl}{\ln 2}\right)$$

4.2.4 Total suspended matter

Total suspended matter algorithms are applied according to the optical water type of the pixel. The selection of the algorithm for each OWT is shown in Table 6.

Table 6. Algorithm selection and optimisation strategy for TSM retrieval per optical water type (OWT)

OWT number	Algorithm source	Algorithm optimisation
3, 5, 9, 11, 13	Based on Binding et al. (2010)	Empirical re-tuning of algorithm parameters based on GloboLakes calibration against the *Limnades* database, specific to each optical water type, analogous to the method described for chlorophyll-a by Neil et al. (2019)
2, 4, 6, 8, 12	Based on Vantrepotte et al. (2011)	
1, 7, 10	Based on Zhang et al. (2014)	

The Binding et al. (2010) algorithm, as it is implemented here, is based on the analytical inversion of reflectance in the 754nm band and converting the resultant particulate backscattering signal using a mass-specific backscattering coefficient for TSM.

The Vantrepotte et al. (2011) algorithm, as it is implemented here is similar but uses the 665nm band. An additional empirical scaling factor was introduced to tune the algorithm to provide the best match for the corresponding water types.

The Zhang et al. (2014) algorithm, as it is implemented here, is an empirical relation between the 709nm band and the in situ TSM, with the algorithm coefficients tuned to provide the best match for the corresponding water types.

For backward compatibility, the turbidity product from versions lower than v2.0 can be converted to the TSM product in v2.0 and up, by dividing by a factor 1.17. The new turbidity product is described in the next section.

4.2.5 Turbidity (version 2.0)

Turbidity in product version 2.0 and higher is obtained from the algorithm described in Nechad et al. (2010, 2015, 2016), which provides flexibility in which input waveband is used whilst providing internally consistent results in the form of either TUR or TSM. The wavelength used for each optical type was selected based on performance compared with the GLORIA in situ reference dataset. Due to frequently negative output the 885 nm band was excluded.

There are insufficient turbidity matchup data available to independently validate the turbidity output from the Nechad algorithm across OWTs (Figure 2). However, the algorithm/waveband assignment can be made on the basis of the TSM output of the algorithm, which effectively addresses observation and upstream algorithm biases, leaving only uncertainty in the relationship between turbidity and TSM unaddressed. Within the Nechad algorithm, this is approached as a constant conversion. The conversion factor used here is turbidity = TSM * 1.04 and is based on regression of in situ TSM vs turbidity with outliers (>2.5 standard deviations) removed (Figure 3Figure 3). In reality, the relation depends on average particle refractive index which will covary, to an extent,

with dry weight (TSM). Algorithm coefficients are tuned for each individual OWT, using matchups corresponding to the top 80% of similarity to the respective OWT. To avoid overfitting and preserve the physical meaning of the coefficients, bounds of ±40% around the original coefficients were set during the tuning process. Blending of the algorithm forms (again based on in situ data) is showing stable performance across the TSM value range, as shown in Figure 4.

Table 7. Algorithm selection and optimisation strategy for TUR retrieval per optical water type (OWT)

OWT number	Algorithm source	Algorithm optimisation
1, 2, 5, 9, 13	Nechad et al. (2010, 2015, 2016)	Using 665 nm band. Tuned per optical water type
3		Using 681 nm band. Tuned per optical water type
12		Using 709 nm band. Tuned per optical water type
4, 6, 7, 8, 10		Using 754 nm band. Tuned per optical water type
11		Using 779 nm band. Tuned per optical water type

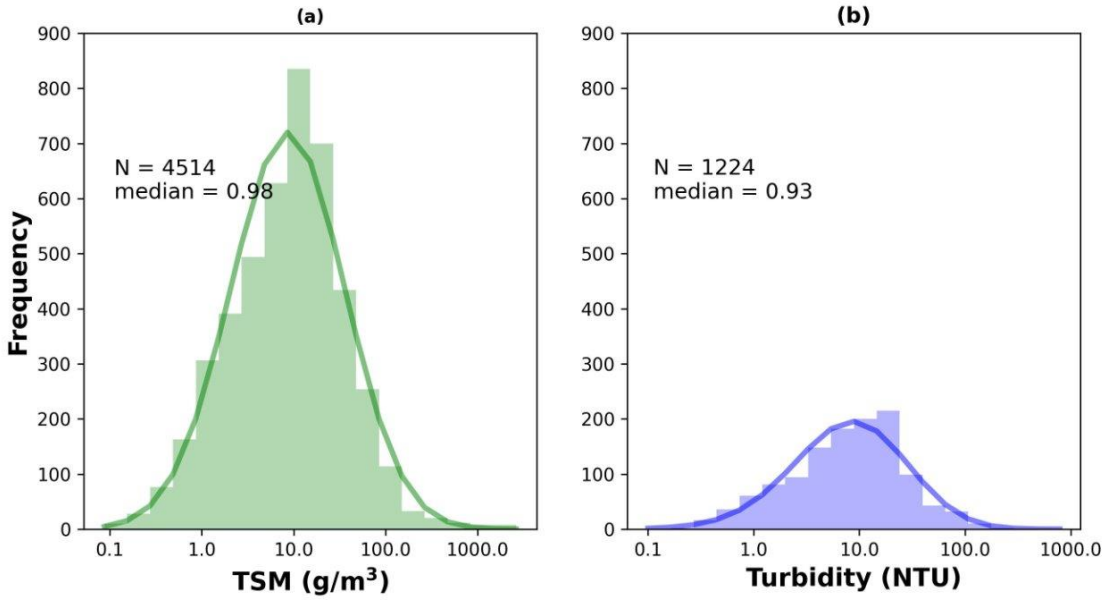


Figure 2. In situ matchups for TSM and turbidity from GLORIA database

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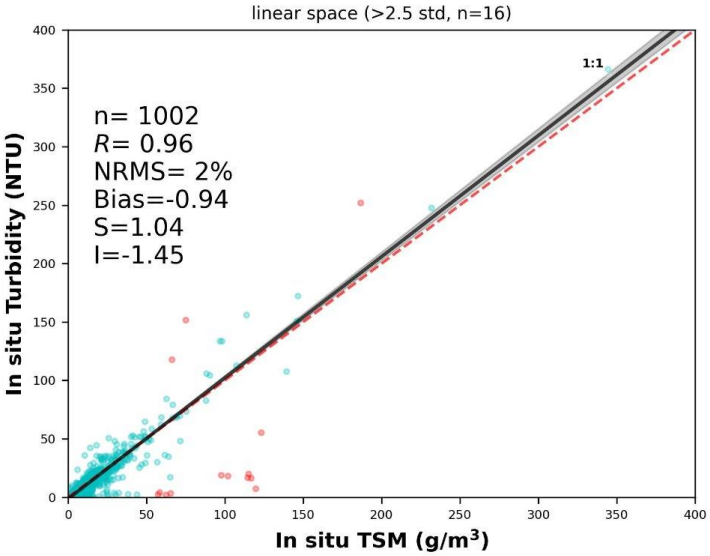


Figure 3. Relationship between in situ TSM and turbidity for the GLORIA dataset. Red points show outliers exceeding 2.5 standard deviations from the mean, that were removed

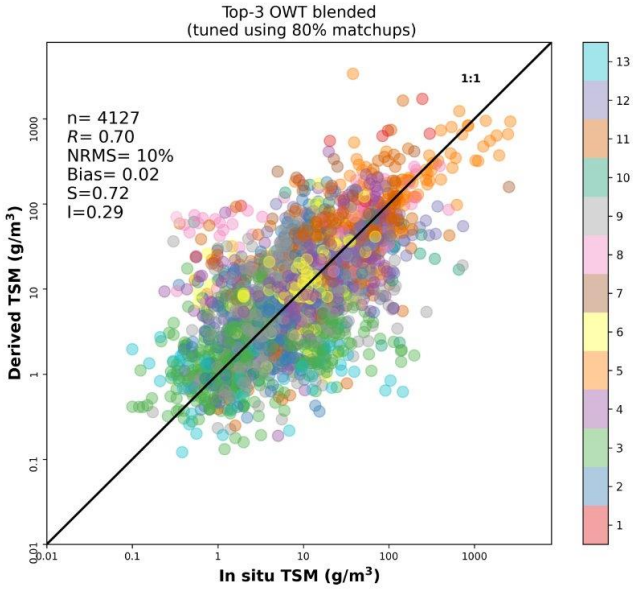


Figure 4. In situ *vs* blended satellite-derived TSM. Colours indicate dominant OWT number

4.2.6 Floating cyanobacteria / floating macrophytes

The Maximum Peak Height (MPH) algorithm from Matthews and Odermatt (2015) is used to detect floating cyanobacteria and floating vegetation. Whereas the original algorithm is based on BRR to partially remove atmospheric effects, the method used here was calibrated and applied to TOA

reflectance, to reduce computation time. The MPH algorithm works on the principle that the concentration of phytoplankton can be estimated from the MPH of the reflectance spectrum in the red to near-infrared region of the electromagnetic spectrum, where the location of the peak is variable. A Sun-Induced Chlorophyll Fluorescence (SICF) peak feature typically appears in low-biomass conditions, centred on the Chl-a fluorescence emission which is observable in 681nm satellite bands and typically more expressed in algal species and vegetation than in cyanobacteria. The Sun-Induced Phycocyanin absorption and Fluorescence (SIPF) feature centres on the spectral band where phycocyanin absorption and fluorescence may appear, which is a characteristic feature of cyanobacteria blooms in nutrient-replete bloom conditions. Comparing these two variables allows cyano-dominant waters to be distinguished from other phytoplankton blooms: cyanobacteria-specific spectral pigmentation features will enhance SIPF and reduce SICF. This work demonstrates how the adapted MPH signatures provide consistent estimates for cyanobacteria surface blooms and floating macrophytes.

The workflow for detection of surface blooms with Sentinel-3 OLCI is shown in Figure 5. First, the normalised difference vegetation index (NDVI) is used to distinguish between a surfacing bloom versus mixed conditions. If a surface bloom is detected, SIPF and SICF are used to distinguish between floating cyanobacteria and floating macrophytes based on defined thresholds.

The following equations for the NDVI, SICF and SIPF are used:

$$NDVI = \frac{B18 - B8}{B18 + B8}$$

$$SICF = B10 - B8 - ((B11 - B8) * \frac{681 - 664}{709 - 664}$$

$$SIPF = B8 - B7 - ((B10 - B7) * \frac{664 - 619}{681 - 619}$$

Exponential functions are used to calculate probabilities of floating cyanobacteria and floating macrophytes using the rate of change of SICF and SIPF with the following equation:

$$-e^{\alpha * SICF - \beta * SIPF} + 1$$

The resulting probability indicates the likelihood of floating cyanobacteria or floating macrophytes presence in each pixel. The weights in the probability function are α = 60 and β = 60.

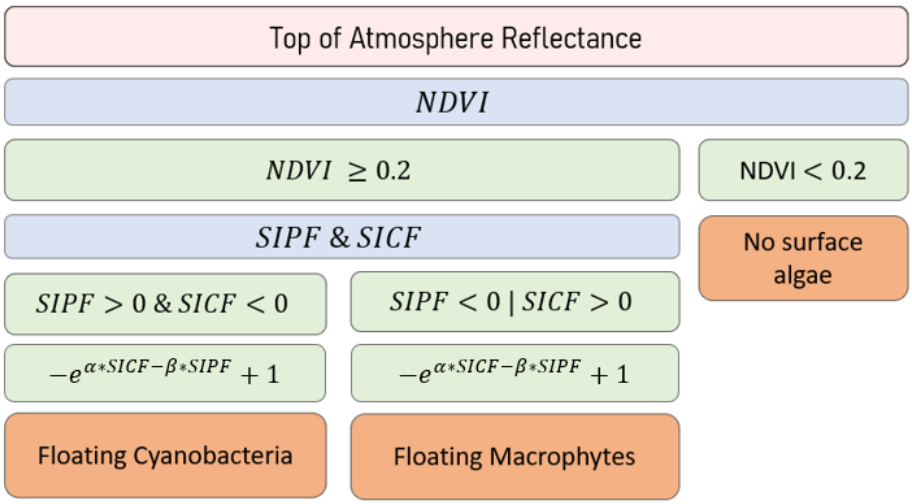


Figure 5. Workflow of detecting surface blooms with Sentinel-3 OLCI

The thresholds used here were identified against visual inspection of processed imagery. Reference data for the actual occurrence of surface blooms and floating aquatic vegetation are generally lacking. Moreover, due to their patchy distribution in time and space, spatial variability of the blooms is not easily captured from in situ observations. The Sentinel 3 OLCI methodology was calibrated against the MPH based on BRR, and was found to yield the same floating biomass indicator values from TOA signals, in imagery where both floating cyanobacteria and floating macrophytes were visually observed.

4.2.7 Per-pixel uncertainty

Per-pixel uncertainty is calculated for chlorophyll-a, TSM and water-leaving reflectance using the method of Liu et al. (2021) as developed for ESA Lakes_cci (Lakes_cci, 2022). The uncertainties associated with the biogeochemical products are propagated from models which take the OWT similarity into account, whereas the uncertainty associated with water-leaving reflectance extrapolates from the global regression of satellite observations against in situ matchups. Future work relating residuals to adjacent land and atmospheric conditions is anticipated but not yet incorporated into the models. Being based on global performance, whilst many in situ reference observations are located close to land, implies that the per-pixel uncertainty likely overestimates in lake centres where remote sensing is the most reliable.

In the water-leaving reflectance uncertainty assessment, systematic negative bias has been repeatedly observed. This bias is likely present in the reflectance observations, but not in the derived biogeochemical products such as chl-a and TSM, because these algorithms have been individually calibrated against in situ reference measurements, cancelling out systematic bias. For this reason, the uncertainty of reflectance products is expressed both as the relative uncertainty (including systematic bias) and the unbiased relative uncertainty. The latter represents the observation and algorithmic uncertainty propagated to the downstream biogeochemical products.

The water-leaving reflectance uncertainty is given for the waveband with the highest reflectance amplitude for each pixel, corresponding to the recommendations in GCOS 244.

4.3 QUALITY ASSESSMENT

4.3.1 Quality Flags

The aggregated 10-day products show NaN if no valid value could be generated. To eliminate invalid or non-water pixels from the aggregation, masks are generated either at processing level 2 (not available to users) or level 3 (the delivered product). These are provided in

Table 8. Where they have been applied, the appropriate flag in the *quality_flags* variable will be raised. Note that the quality_flags defined in Table 3 have definitions that are not included in

Table 8 as those pixels are not excluded, rather the flag is raised for information on quality risks, e.g., when *floating_cyanobacteria* are likely present.

Table 8. Masking of output variables

Description	Condition	Flag generation stage	Generated / Applied at Processing Level
Invalid L1 input	INVALID	*IdePix*	L2 / L3 (all output variables)
Water mask	land_water_fraction < 100 or SNOW_ICE	*IdePix*, land water mask	L2 / L3 (all output variables)
Cloud mask	CLOUD or CLOUD_BUFFER or CLOUD_SHADOW or CLOUD_AMBIGUOUS or CLOUD_SURE	*IdePix*	L2 / L3 (all output variables)
Bright pixel	$R_w412 > 0.4$ or $R_w865 > 0.4$	Polymer	L3 (R_w variables are not masked)
Extreme TUR value	Turbidity > 100000 or Turbidity < 0 NTU or Turbidity = Inf	Aggregation	L3 (turbidity)
Extreme Chla value	Chla < 0 or Chla > 10000	Aggregation	L3 (chl-a, TSI)
Extreme TSM value	tsm < 0 or tsm > 10000	Aggregation	L3 (TSM)

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Description	Condition	Flag generation stage	Generated / Applied at Processing Level
Land adjacency	(owt_14 + owt_15>1.59) and ((owt_dominant_class==14) or (owt_dominant_class==15))	OWT classifier	L3
Polymer out-of-bounds	Raised by Polymer if bio-optical model parameters exceed prescribed limits	Polymer	L2 / L3 (all output variables)

4.3.2 Algorithm performance

4.3.2.1 Cloud screening

The validation of the cloud identification has been performed with the *PixBox* validation, which compared manually selected pixels against the CLOUD, CLEAR and SNOW/ICE flag computed with *IdePix*. The reference dataset (*PixBox* dataset) holds manually selected pixels that are labelled according to their cloud category. Pixels over different water types are included. CLOUD flag and references are compiled in a confusion matrix showing the percentage of agreement, producer's and user's accuracy (Figure 6).

The overall accuracy reaches 90%, while the user accuracy over water is the best (99.2%) against 81.3% Producers Accuracy. This means that *IdePix* tends to overestimate cloud pixels and even classifies clear water pixels as cloud. The accuracy over ice pixels is lower, which reflects the difficulties to separate between cloud and ice surfaces.

		Pixel Identification				
		WATER	CLOUD	SNOW_ICE	∑	Producers Accuracy
PixBox	Clear water	6712	1096	446	8254	81.3
	Cloud	29	16208	279	16516	98.1
	Clear snow	27	807	1209	2043	59.2
	∑	6768	18111	1934	26813	
	Users Accuracy	99.2	89.5	62.5		**90.0**

Figure 6. Confusion matrix showing the classification accuracy of *IdePix* clear water, cloud and snow/ice detection

4.3.2.2 Atmospheric correction

4.3.2.2.1 Atmospheric correction algorithm selection

An earlier version of Polymer (v3.5) was validated against in situ reflectance data contained in *Limnades* within the UK NERC GloboLakes project in a round robin matchup exercise of six candidate algorithms. Polymer gave the best systematic performance (unbiased error) in each waveband, although a consistent underestimation was apparent. This systematic error is cancelled out in the whole-chain calibration of TSM, turbidity and chlorophyll-a retrieval. A correction is not attempted for the reflectance because there are not enough in situ validation data available to inspect the performance per OWT or for the wide variety of observation conditions (adjacent land, bottom effects, atmospheric composition) that may be encountered. Further improvement of the Polymer atmospheric correction has been based on results of ongoing investigations, which have thus far shown that a mineral absorption model should not be universally applied, as was previously considered a solution. Instead, a wider range of algorithm initialisation conditions has shown that a wider range of output amplitudes can be produced by the algorithm, facilitating retrieval of turbid water types. This upgrade is staged for product v3.0 as the required recalibration of biogeochemical retrieval algorithms is still in progress. It should be noted that underestimation of reflectance is not fully propagated to water constituent retrieval algorithm performance as long as the shape of the spectrum remains similar, because systematic bias is addressed in downstream algorithm calibration.

Polymer v4.14, as used in v2.0 of the lake water quality products has seen several improvements since v3.5:

- Version 4 and beyond are written in Python, v3.5 is written in C++
- Since v4.6, Polymer includes an option to account for mineral absorption. Evaluation against in situ measurements has shown this is not a good solution to address the negative bias
- v4.6 and higher support Sentinel-3A/B OLCI and Sentinel-2A/B MSI data
- MERIS results for v4.6 (without “mineral absorption” option) and v3.5 are close to identical
- v4.8 takes elevation into account
- With v4.12, a significant update of SVCs for Sentinel-3A OLCI has been applied (same gains used for S-3B)
- Improved stability in the sunglint
- Support for ERA5 ancillary data has been added
- The spectral configuration for OLCI is adjusted to avoid occasional unstable behaviour (Band 412 is not used for atmospheric correction anymore but written to the output product) (v4.11)

Version 2.0 of the Lake Water Quality processing uses Polymer v4.14. The previous LWQ product versions 1.3 and 1.4 used Polymer v4.12. As used in this processing chain v2.0, there are no significant differences in the Polymer versions, and a negative bias is still apparent and represented in the uncertainty products. Reflectance data from versions 1.3, 1.4 and 2.0 can be considered equivalent. The performance of v4.14 compared against in situ data from the GLORIA dataset (Lehmann et al. 2023) is shown in Figure 7.

The full change log can be read on the software website: https://www.hygeos.com/static/polymer/CHANGELOG.TXT

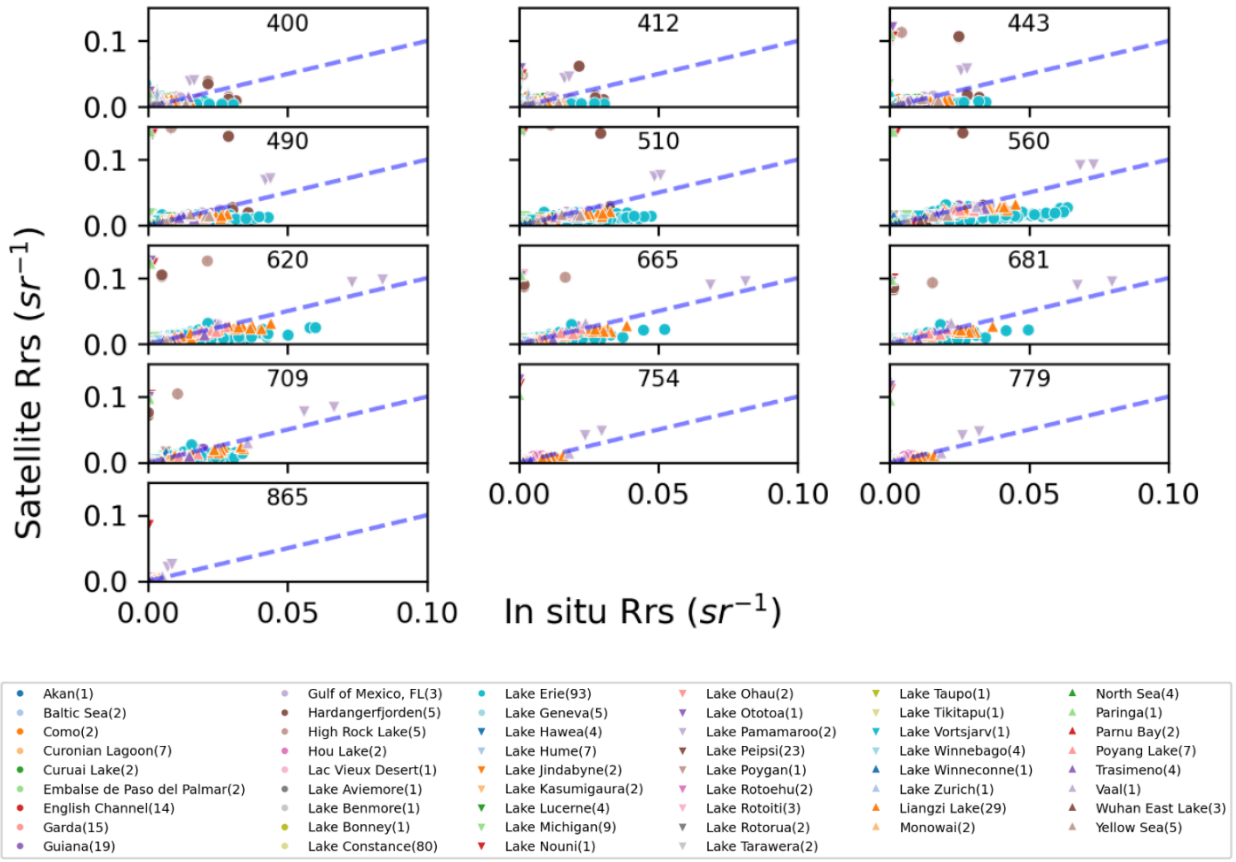


Figure 7. Matchups of OLCI data processed with Polymer v4.14 compared with the GLORIA in situ dataset

4.3.2.3 Water constituent retrieval – matchup analysis

The chlorophyll-a concentration (from which Trophic State Index is also derived) can be compared against in situ data. Figure 8 shows the results of the comparison using the chlorophyll-a algorithms associated with the most similar Optical Water Type, as performed in the UK NERC GloboLakes project using in situ concentration and in situ reflectance data. This is shown for reference as the best attainable result (as atmospheric correction bias is excluded) when using satellite imagery with dynamic algorithm selection but without algorithm blending procedures. The scatterplots show the chlorophyll-a concentration separated by optical water types (in different colours) and corresponding chlorophyll-a algorithms (in different marker shapes).

The dataset of MERIS satellite matchups using the *Calimnos* set of algorithms, against any available corresponding in situ observations in 2002-2012, is notably smaller and necessitates validation over a ±7-day matchup window to have sufficient data for comparison across water types. These results, shown in Figure 9, do not cover all water types (no results where OWTs 1,

10, 11, 12, or 13 were dominant), and due to the longer matchup window should be interpreted as the worst case because environmental conditions are likely to change over several days. These results show a slight systematic underestimation, and a coefficient of determination $R^2 = 0.62$ over the observed range (0-70 mg/m^3).

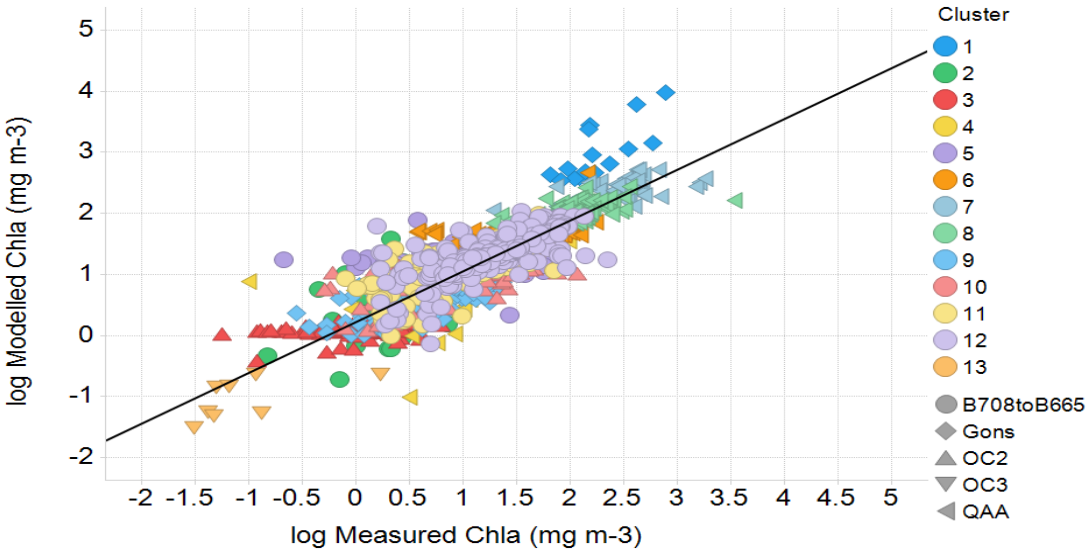


Figure 8. Performance of chlorophyll-a retrieval across all optical water types, OWTs (clusters 1-13) and associated algorithms following tuning of each algorithm, for each OWT, against the *Limnades* database (results courtesy University of Stirling, UK NERC GloboLakes, Neil et al. 2019)

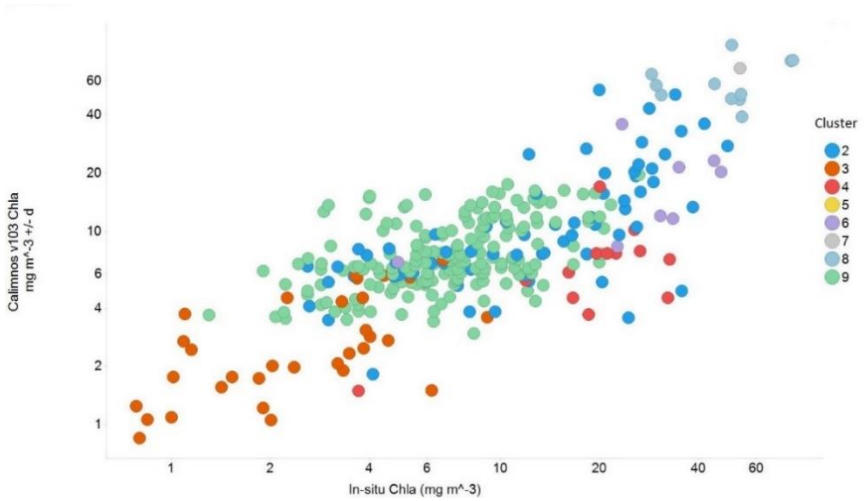


Figure 9. Satellite matchup analysis for chlorophyll-a retrieval in the *Calimnos* MERIS processing chain. Linear regression analysis provides $R^2 = 0.62$, slope = 0.82, intercept = 1.16, n = 350. To obtain sufficient matchup points, a matchup window of ±7 days was used. Temporal variation in this

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timeframe can be significant, contributing to scatter in the observed relationship. The result should be interpreted as worst-case

A more recent uncertainty characterisation analysis using the *Calimnos* v1.04 dataset shows the improvement of the blended chlorophyll-a result over individual algorithms (Figure 10). The most suitable lake-wide algorithm (left panel) was allowed to vary per observation day (which is already an improvement over applying the same algorithm every day). The per-pixel blended chlorophyll-a (here shown using just the top-2 highest ranking OWT-algorithm combinations) clearly has a higher dynamic range and associated linearity. This can likely be attributed to the occurrence of smaller patches of phytoplankton bloom with extreme concentrations. The dynamic algorithm selection approach is, therefore, preferred as it is better equipped to deal with optical gradients within individual lakes.

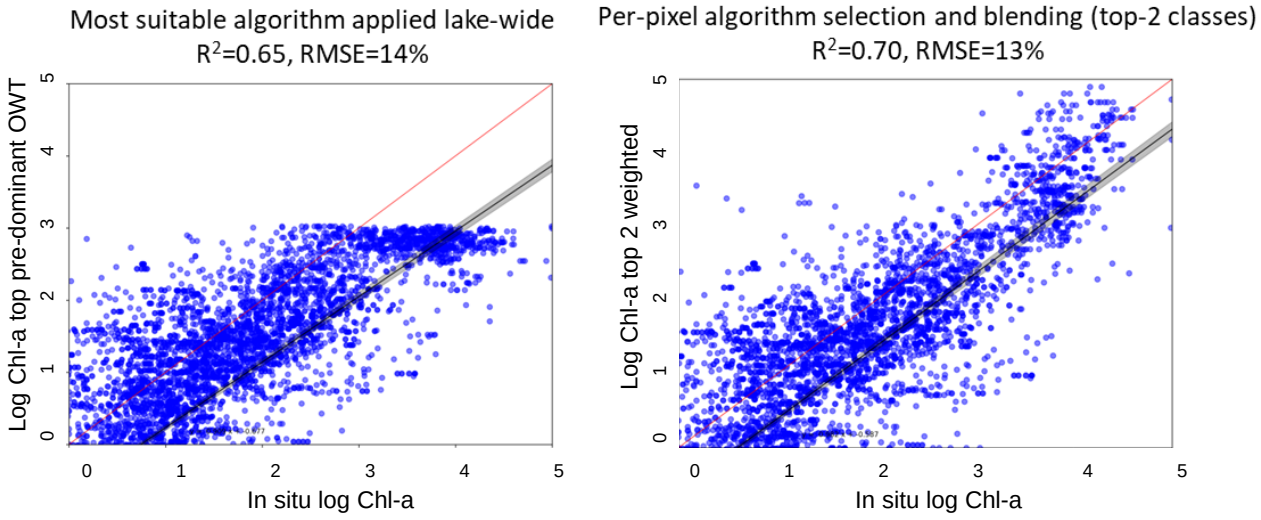


Figure 10. Comparison of the performance of (left) the best chlorophyll-a algorithm for the predominant optical water type of each lake and (right) per-pixel selection and blending of the two highest ranking algorithms based on optical water type membership scores

The final, blended chl-a and TSM algorithms included in v2.0, using MERIS as input, compare against the LIMNADES in situ data archive as shown in Figure 11.

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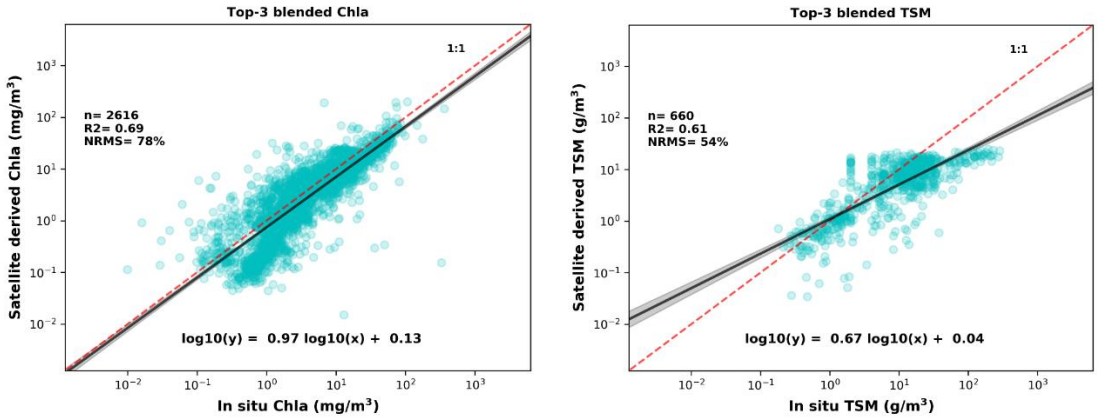


Figure 11. Comparison between in situ and top-3 blended (left) chlorophyll-a and (right) TSM derived from MERIS

4.3.2.4 Water constituent retrieval – time series of water quality parameters

Time series comparison between in situ data and EO derived parameters enable to assess the behaviour of both measurement techniques over time. The focus is on the consistency of the time series on the one hand and on the comparability of the datasets on the other hand. The order of magnitude and seasonal patterns are investigated. Two time series are shown for Lake Michigan, for which recent in situ data were available via the US databases STORET (http://www3.epa.gov/storet/) and WQP (http://waterqualitydata.us/portal/).

Lake Michigan is dominated by clear water falling into OWTs 13 and 3. Only close to the coast more turbid waters can be found. For demonstration, two time series are shown for a station in the clear lake water and for one station close to the shore. They are marked in Figure 12 with two pins. The respective time series are shown in Figure 13. The upper plot shows the time series in the central part of the lake with turbidity below 0.8 NTU, the lower plot for the coastal station shows turbidity up to 5 NTU. In situ data (orange points) and the CGLOPS turbidity product (blue crosses) correspond in their magnitude and in some parts in the seasonal cycle. As the CGLOPS turbidity product is a 10-day average, differences are within expectation in the more dynamic area like the coastal station near the river inflow.

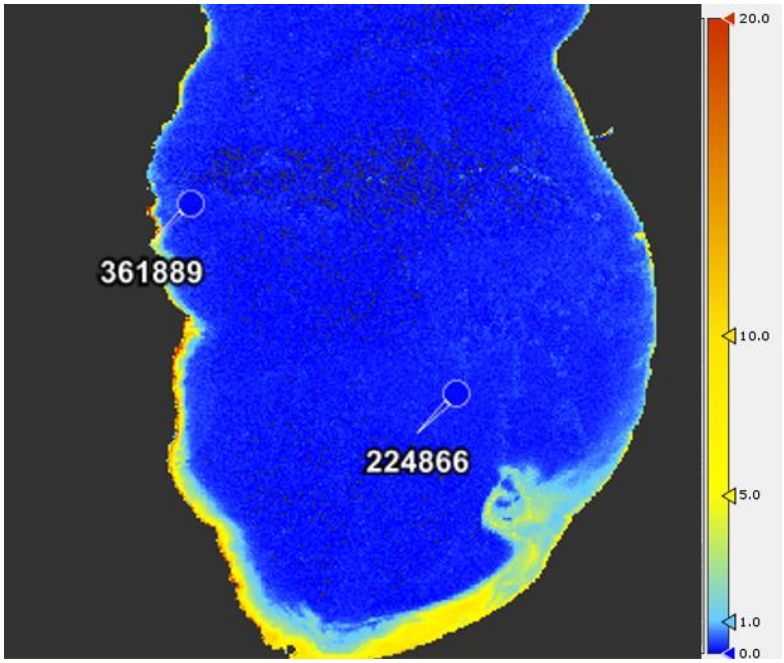


Figure 12. Turbidity in Lake Michigan (10-day average 20170401-20170410, S-3 OLCI). Pins indicate the positions shown in the time series plots

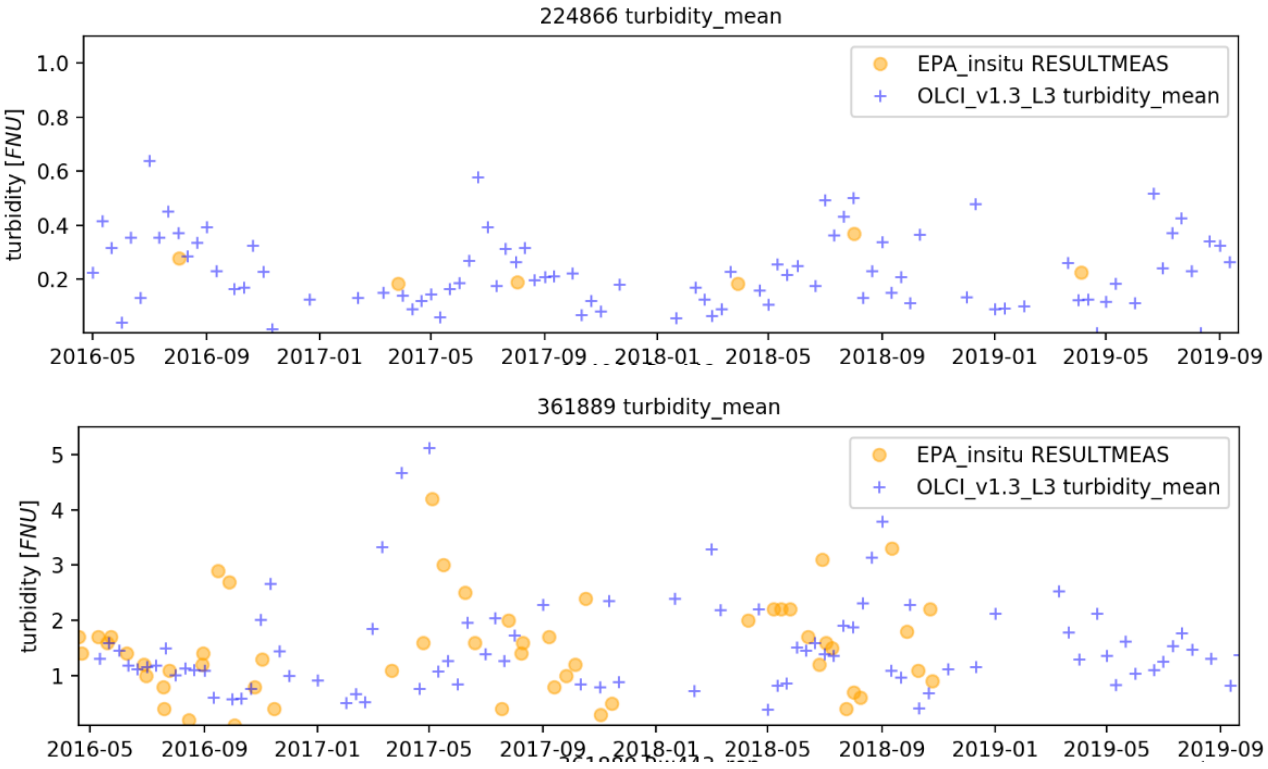


Figure 13. Turbidity time series at two stations in Lake Michigan. Clear waters in central part of the lake (upper panel) and more turbid water close to the coast (lower panel). In situ data: US databases STORET (http://www3.epa.gov/storet/) and WQP (http://waterqualitydata.us/portal/)

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An example for the comparison of the Trophic State Index and in situ chlorophyll-a concentration measurements, sampled for monitoring purposes, is shown in Figure 14 for Lake Vänern (Sweden). The chlorophyll-a measurements were transformed to the TSI categories used for the CGLOPS products. The station indicated by the red arrow is shown in the upper time series plot (Figure 15) which shows a majority in TSI of 50 in both datasets, while the second station (blue arrow, lower time series plot, Figure 15) aligns for both datasets at a TSI of 70. The first example shows some more variability in the satellite TSI class than the in situ data.

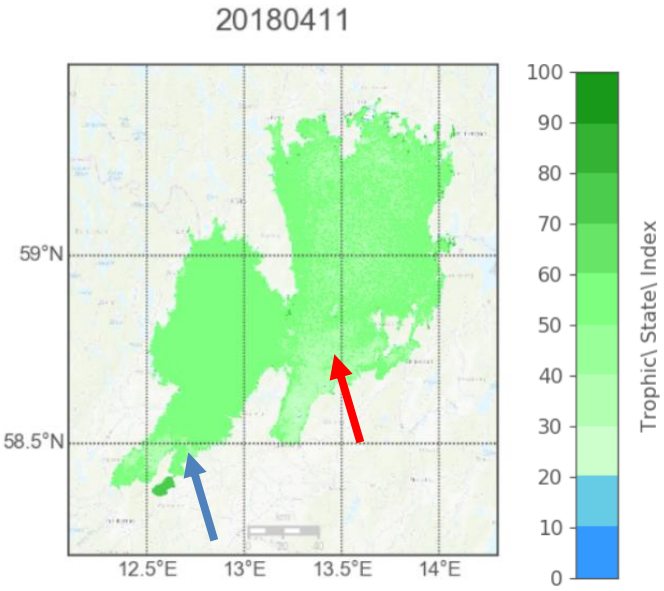


Figure 14. Lake Vänern – Trophic State Index for the second decad in April 2018 (OLCI). The arrow indicates the position of the stations shown in Figure 15 (red arrow for upper panel, blue arrow for lower panel)

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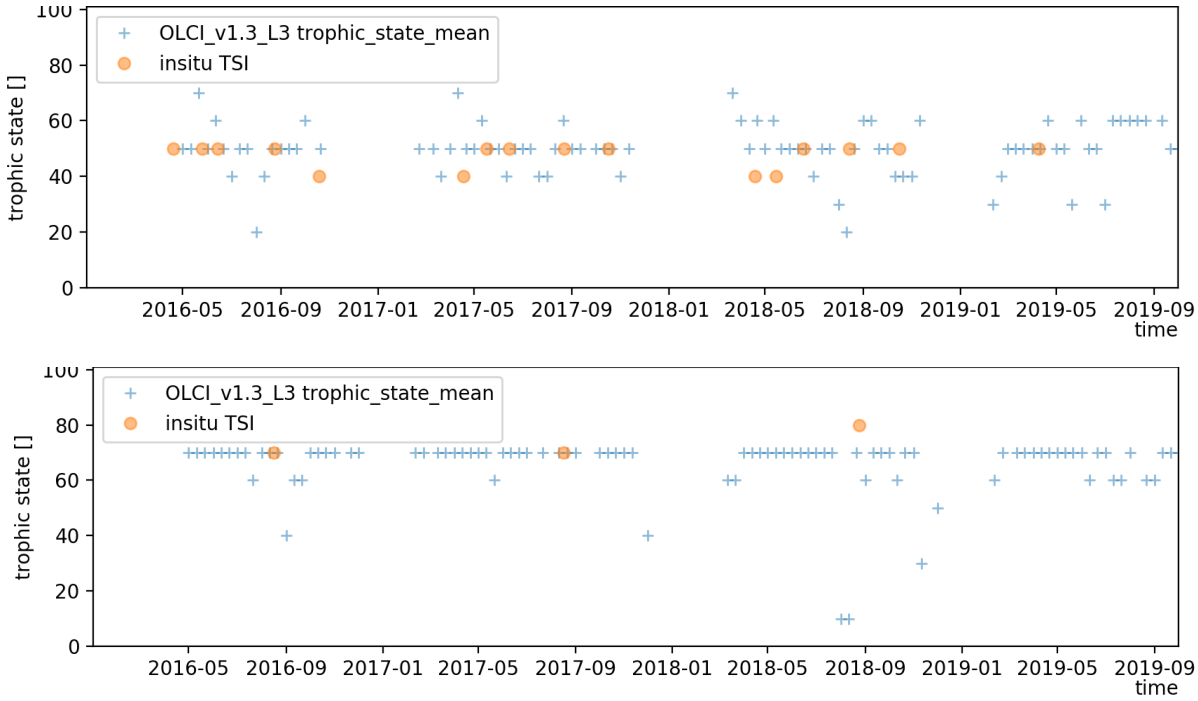


Figure 15. Time series in Lake Vänern showing the consistency with trophic state based on in situ chlorophyll-a concentration. The in situ data is collected within the framework of the coordinated Swedish environmental monitoring programme

4.3.2.5 Floating cyanobacteria and floating macrophytes

Two approaches are used to verify the accuracy of identification of floating cyanobacteria and floating macroalgae. In situ measurements are scarce. Consequently, we concentrated on visual inspection of the spectral behaviour within the floating blooms and their surroundings as well as consulting online media and bloom reports of individual events.

An example for the spectral assessment is given for Lake Erie. Lake Erie is known to exhibit surface blooms of cyanobacteria in summer, particularly in the western parts. Here, we look at a profile of spectra from open waters to those influenced by riverine inputs (Figure 16). Two locations within the floating cyanobacteria mask correspond to highly elevated near infra-red TOA radiance, whereas surrounding waters are representative of mixed conditions. In the visible domain, the spectra have similar shapes including a sharp drop from 560nm to 620nm, which is often associated with phycocyanin absorption. Elevated reflectance is visible at all locations within the bloom, which shows a clear front between pins 5 and 6. Atmospheric correction is again shown to fail in the area affected by floating cyanobacteria, and the results are also not stable, with similar TOA spectra resulting in large difference in shape and amplitude of the corrected spectra.

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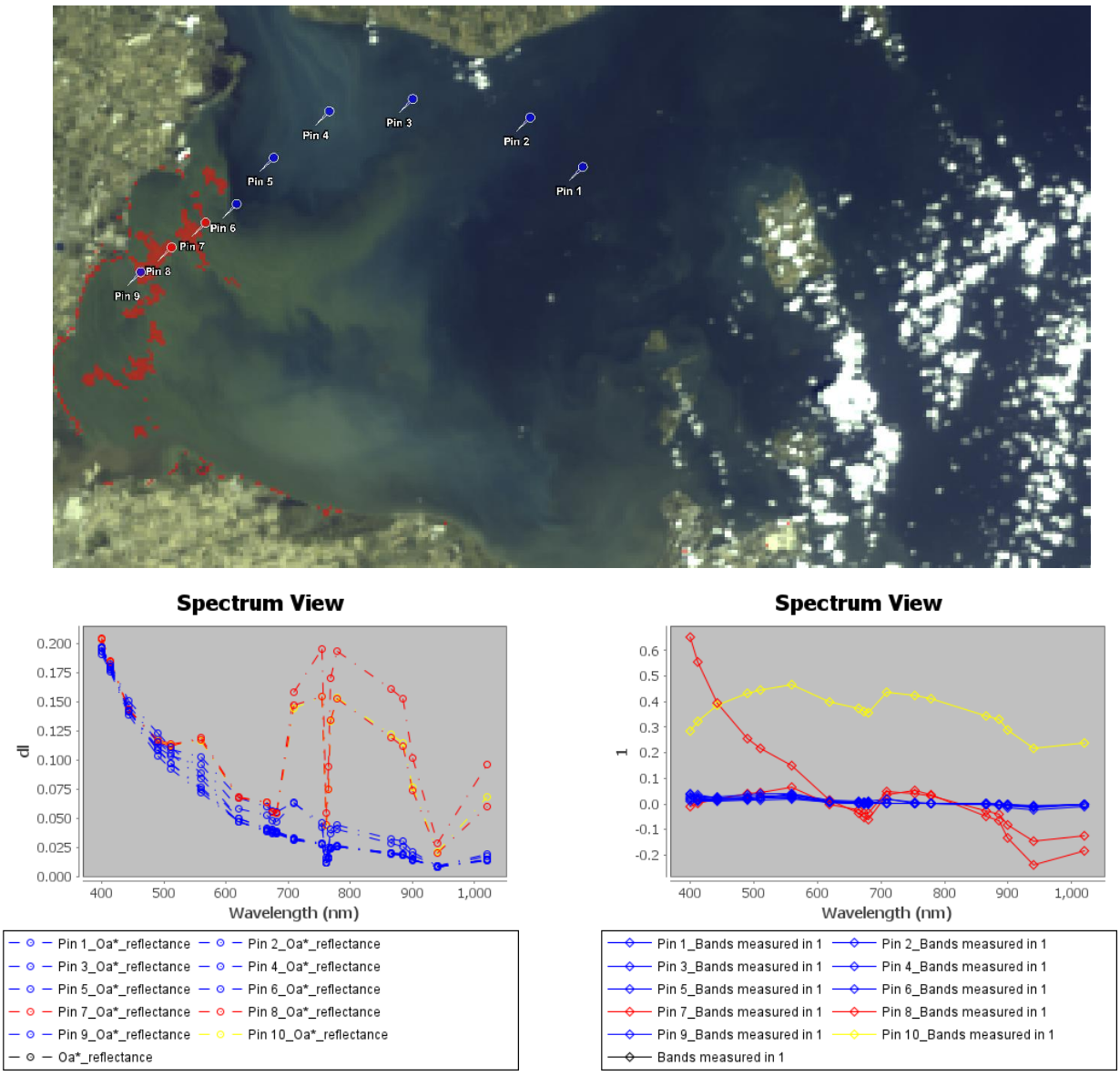


Figure 16. Floating cyanobacteria in western Lake Erie. (top) RGB image from TOA reflectance with floating cyanobacteria mask overlaid in red, and the locations for which spectra are plotted in the bottom graphs. (bottom-left) TOA reflectance spectra, (bottom-right) atmospherically corrected spectra

The distinction between floating cyanobacteria and floating macrophytes is demonstrated for a blooming event in Winam Gulf. The Winam Gulf is a large bay in the north-eastern part of Lake Victoria. It is approximately 130km long and 30km wide, located in Kenya. Winam Gulf is known to have abundant floating macrophytes and cyanobacteria. Floating macrophytes provide habitat for various species of fish, insects, and other aquatic organisms. In Winam Gulf, some of the common floating macrophytes include water hyacinth, water lettuce, and papyrus. Blooms of cyanobacteria

have been observed in the Winam Gulf, and they can cause problems for local fishing communities who rely on the lake for their livelihoods.

Figure 17 and Figure 18 show an example of the detection of floating material, with S-2 MSI and S-3 OLCI. Both images show similar spatial patterns for the abundance of floating macrophytes and floating cyanobacteria in the same bay. The EO images were taken three days apart and it seems that the macrophytes drifted west over that time. The cyanobacteria surface blooms seemed to be locally stable in the south-western part of the bay. Visual inspection showed that the methodology worked well, based on the spatial pattern of each type of floating blooms. The spatial pattern of floating cyanobacteria shows wind-swept filaments whereas the floating macrophytes are more strictly clustered. Several articles confirmed the presence of water hyacinth and cyanobacteria blooms in Winam Gulf.

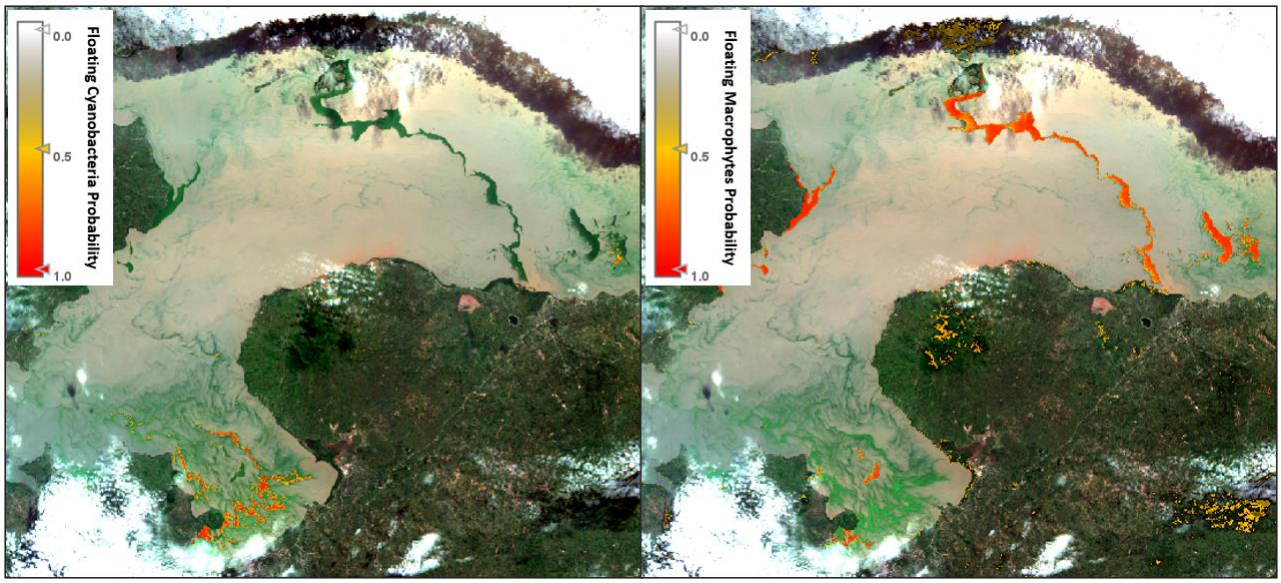


Figure 17. Detection of floating cyanobacteria (left) and floating macrophytes (right) in Winam Gulf with Sentinel-2 MSI on 26-06-2018

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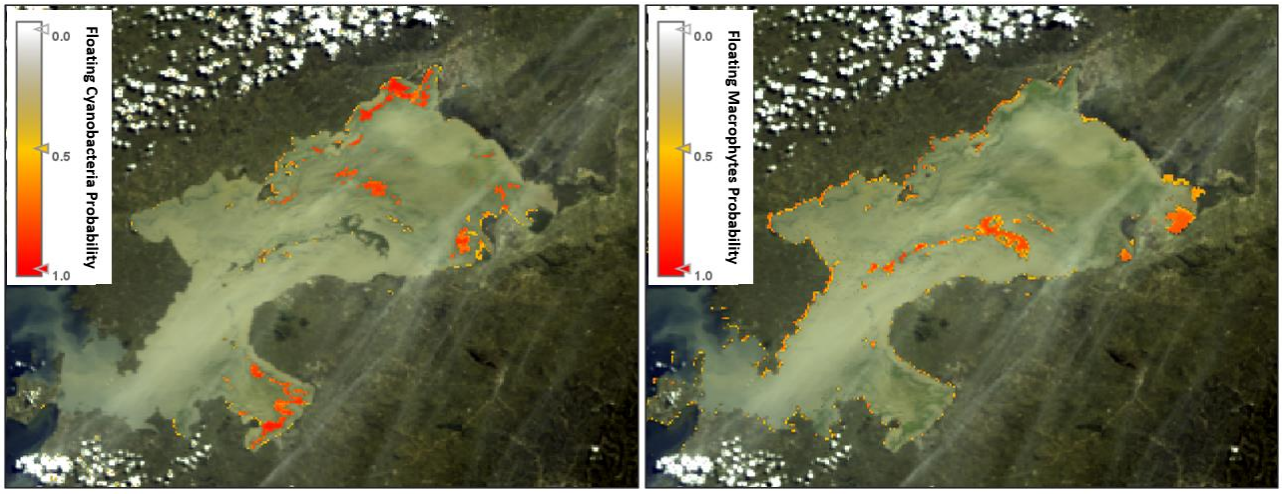


Figure 18. Detection of floating cyanobacteria (left) and floating macrophytes (right) in Winam Gulf with Sentinel-3 OLCI on 23-06-2018

4.3.3 Performance of mapping per optical water type

The optical water type itself is not a parameter within the final LWQ products delivered to the users. However, some examples of their distribution and the influence they have on the blending of algorithms shall be demonstrated here. Maps of the OWT in two lakes as well as spatial transects of turbidity are provided in Figure 19 up to Figure 22. It was investigated whether changes of classes introduce spatial discontinuities into the parameters. Neither the image nor the transect in Lake Huron (USA) show transitions at class borders, however the algorithm for turbidity is the same for the three identified OWT classes 3, 9 and 13.

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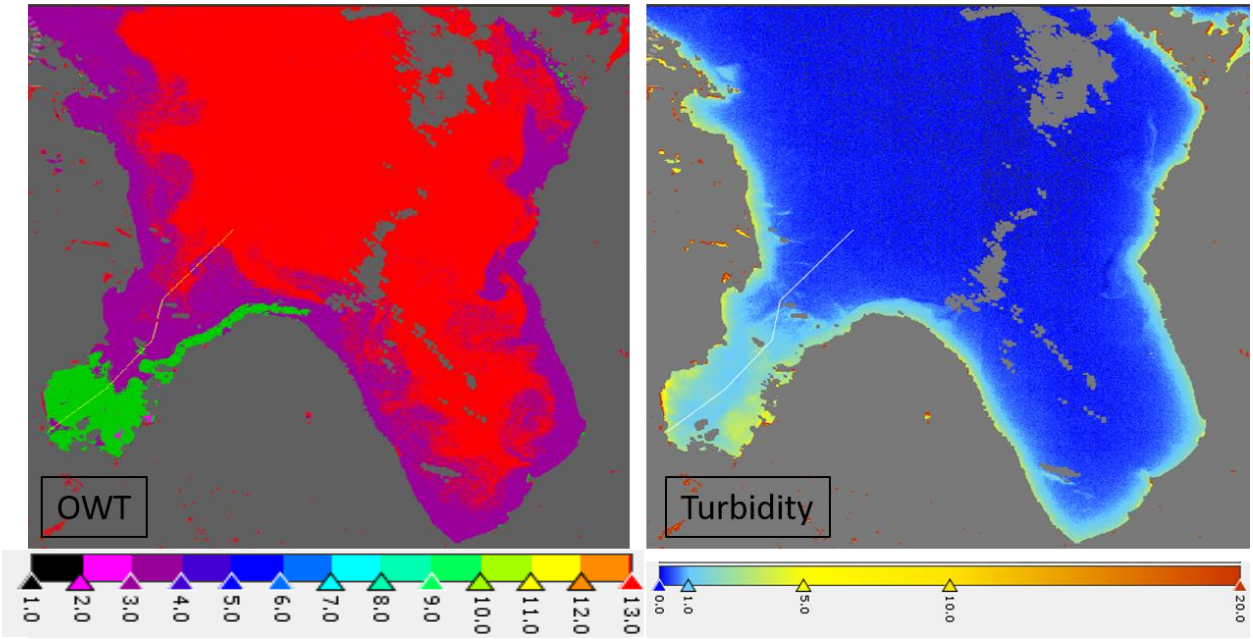


Figure 19. Lake Huron OWT (left) and Turbidity in NTU (right) for Sentinel-3 OLCI on 26-08-2017. The line indicates the transect shown in Figure 20

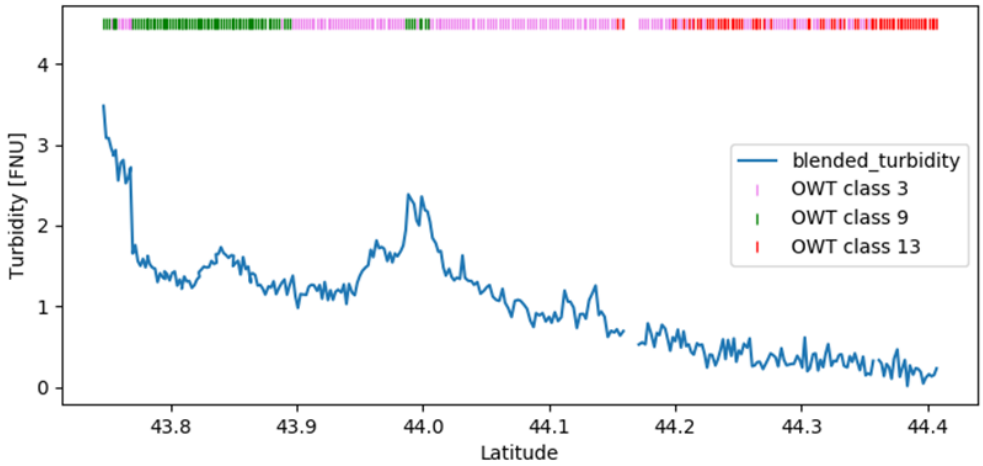


Figure 20. Blended turbidity and OWT classes along a transect in Lake Huron (position of transect in Figure 19)

Figure 21 and Figure 22 show the same setup for Lake Turkana, which is characterised by OWT classes 2, 4, 9 and 11. The small peaks in the turbidity correspond to single pixels of OWT class 9 inside larger areas of OWT class 2. No transition in the turbidity is visible between the OWT classes 4 and 9 although different algorithms are applied for OWT dominant classes 2 and 4 than for OWT dominant classes 9 and 11.

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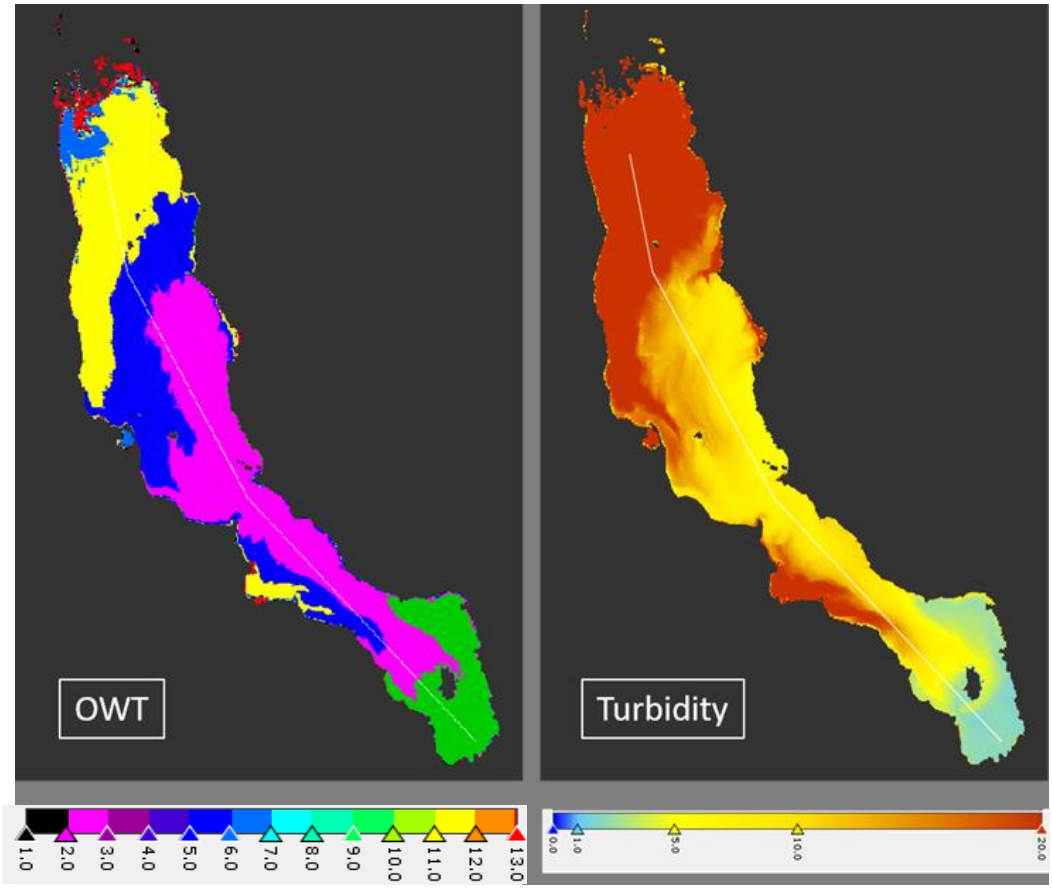


Figure 21. Lake Turkana OWT (left) and Turbidity in NTU (right) for Sentinel-3 OLCI on 19-09-2017. The line indicates the transect shown in Figure 22

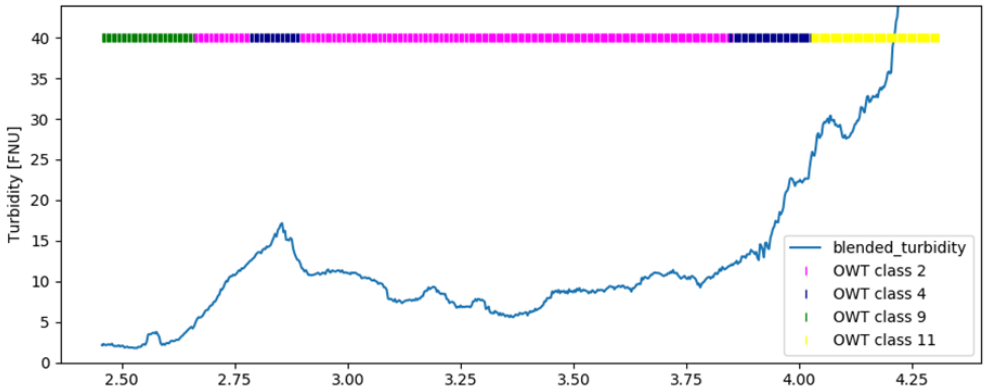


Figure 22. Blended Turbidity and OWT classes along a transect in Lake Turkana (position of transect in Figure 21)

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4.3.4 Consistency assessment

Selected lakes are checked for consistency in the spatial value distribution, for the same dekads in different years. The first example is shown in Figure 23 for Lake Huron. TSI is shown for the three dekads of May in years 2017, 2018 and 2019. The central lake is characterised by the lowest trophic state (10) while trophic state increases towards the shorelines and in Saginaw Bay (large south-western Bay of Lake Huron). The patterns are similar in all three years.

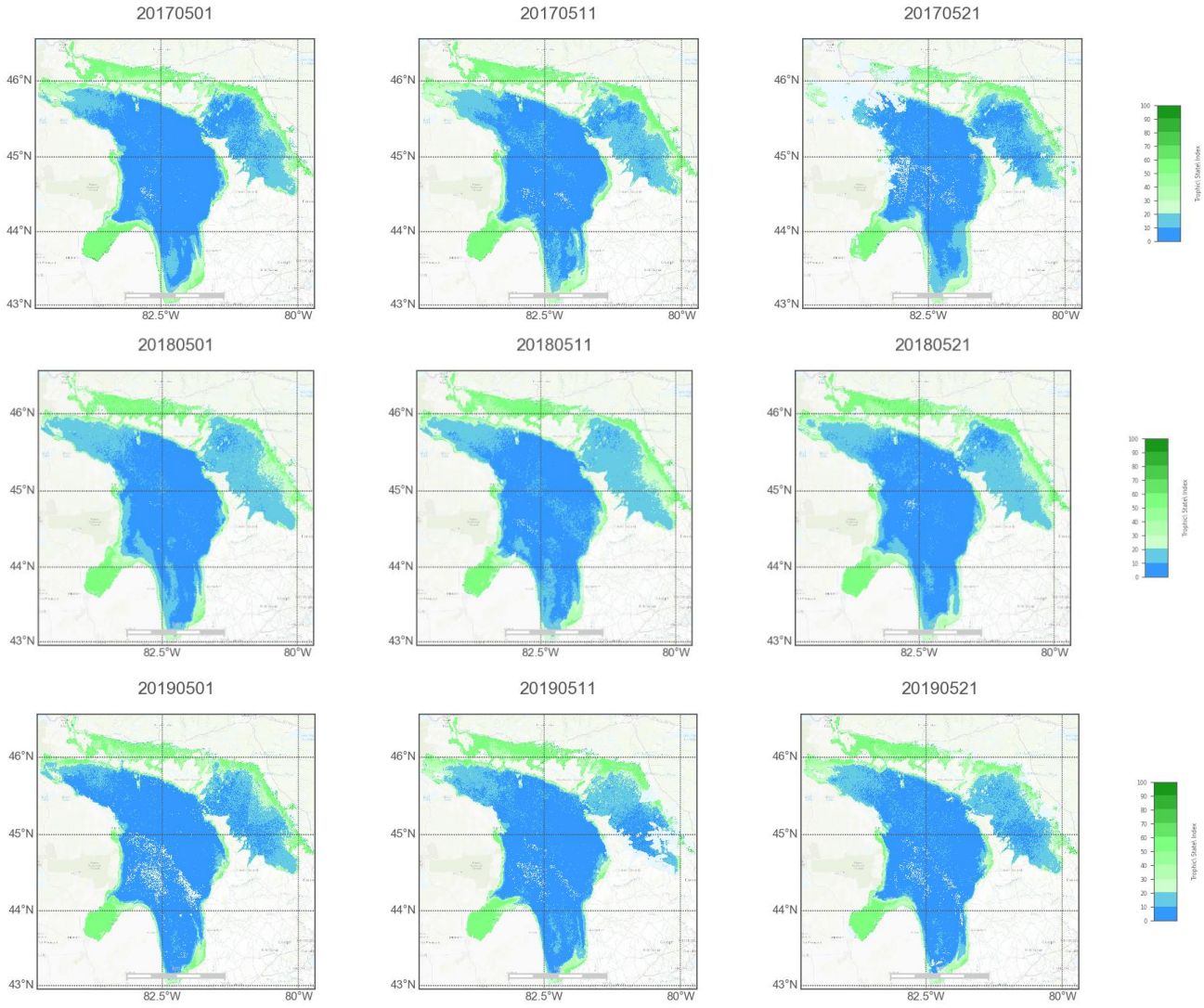


Figure 23. Trophic State Index for the three dekads of May in Lake Huron. Top panel: 2017, middle panel: 2018, bottom panel: 2019

A second example demonstrates the different levels of turbidity in Lake Turkana during the dry and wet seasons. Two dekads are chosen for March (dry season) and August (rainy season). During the rainy season, river discharge is larger and suspended sediment is transported towards and into the lake. The effect is predominant in the northern part of the Lake (River Omo), but in smaller

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scale also visible in the middle-west of the lake, where River Turkwel introduces sediments into the lake.

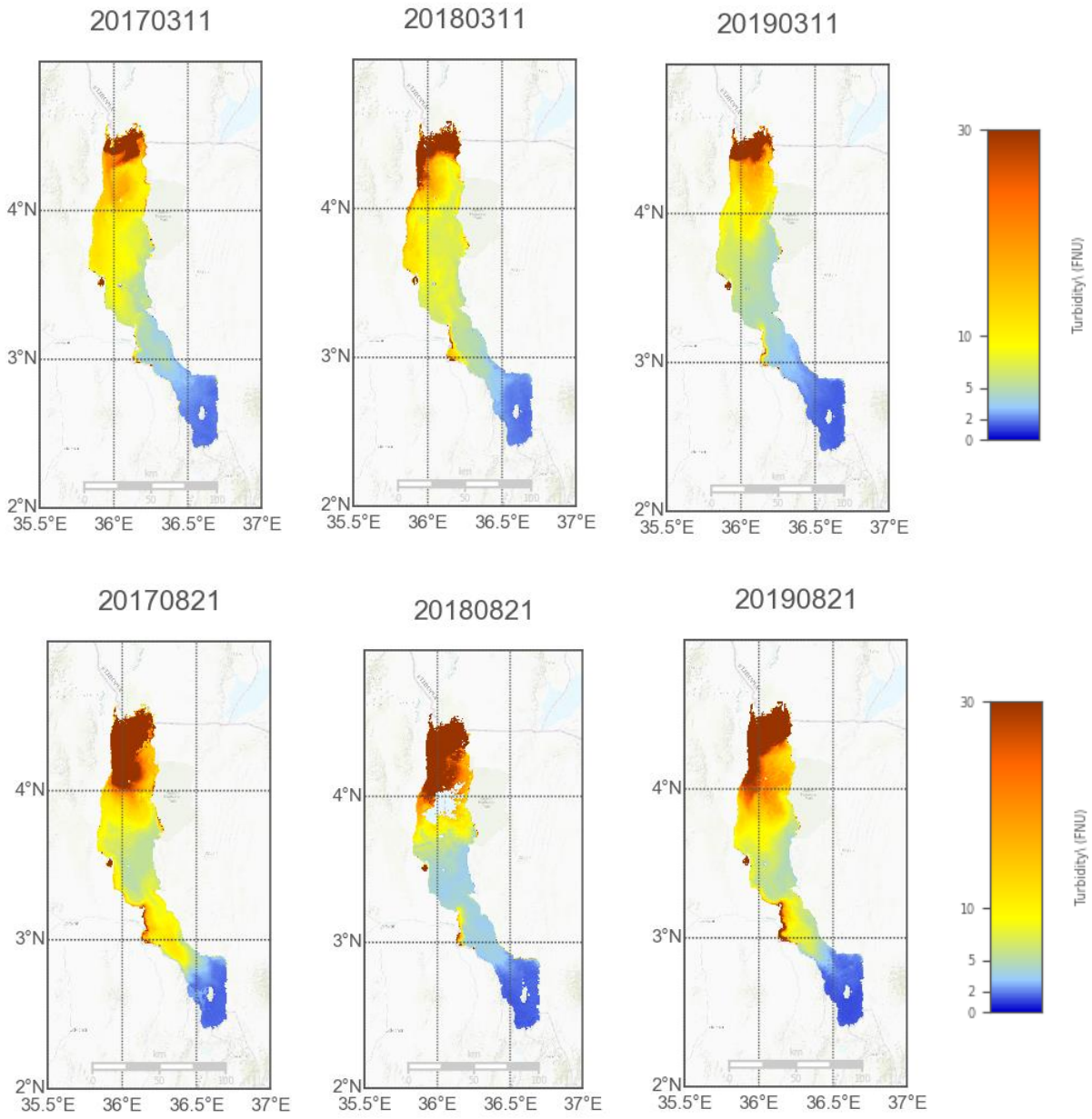


Figure 24. Turbidity in Lake Turkana for two different dekads in 2017, 2018 and 2019 – top panel: dry season (less river inflow), bottom panel: wet season (strongly influenced by river inflow)

4.3.5 Addressing artefacts in version 2.0 and 3.0

Artefacts have been observed in the most turbid lakes in the dataset, e.g. Lake Tana (Figure 25b). By applying the Polymer out-of-bounds flag (Figure 25a) affected regions can be removed. This is

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effective but data are lost (Figure 25c). This additional flag is applied from product version 2.0 onwards.

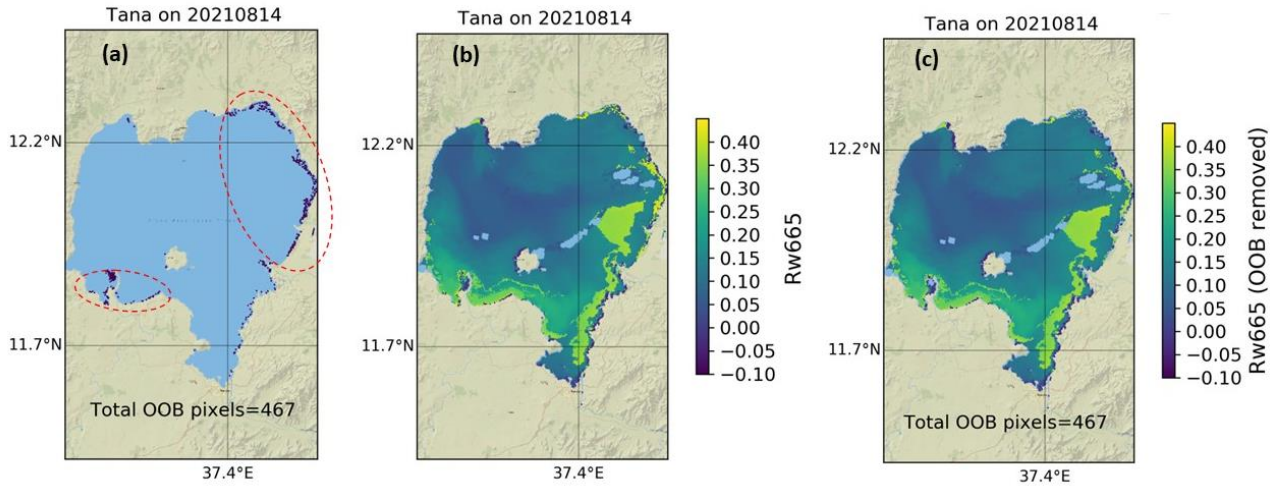


Figure 25. Maps of (a) out-of-bounds from Polymer, (b) R_w665 from lake water product v1.4, and (c) R_w665 v2.0 in lake Tana on 14-08-2021

A new version of Polymer (v4.15 and higher) has been released in response to this issue. A range of initialisation conditions are tried for each pixel and the mineral absorption model is now applied based on a log_chl threshold rather than being permanently on or off. The new versions are effective in tackling the high turbidity artifacts (Figure 26) and are considered the candidate atmospheric correction algorithm for future LWQ product v3.0 (expected 2025).

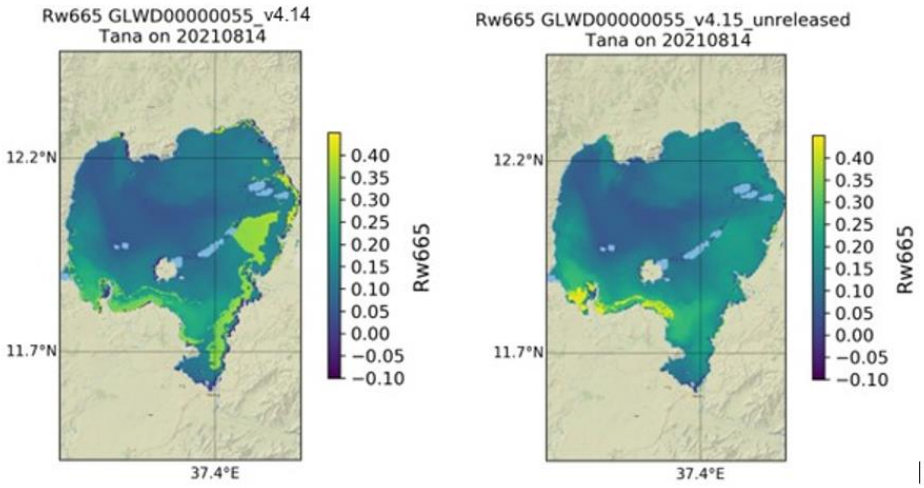


Figure 26. Maps of R_w665 from Polymer v4.14 and v4.15 in lake Tana

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